# *Reducing Italy's Energy Dependence: An Economic Comparison of Renewables, Energy Efficiency, Electrification and Grid Flexibility*

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## *Energy and Environmental Economics*

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# *Index*


***1. Introduction*** ..... 4

***2. Theoretical Framework: Energy Security, Energy Accounting and the Energy Trilemma*** ..... 5

***3. The Italian Case: The Current Structure of the Energy System*** ..... 7

3.1 Import Dependence and Primary Energy Supply ..... 7

3.2 Final Energy Consumption: Where Energy Is Actually Used ..... 7

3.3 Sectoral Exposure: Transport, Buildings and Industry ..... 8

3.4 The Electricity System: Cleaner, but Still Strategically Central ..... 9

3.5 Main Structural Criticalities of the Italian System ..... 10

3.6 What the Italian Case Implies for the Research Question ..... 10

***4. Methodology: A Multi-Criteria Assessment of Energy Dependence Reduction Options*** ..... 12

***5. Energy Efficiency: Reducing Demand Before Replacing Supply*** ..... 15

5.1. Economic rationale ..... 15

5.2. Why energy efficiency matters for Italy ..... 15

5.3. Policy instruments and the “failure-matching” principle ..... 16

5.4. Limits and trade-offs ..... 16

5.5. Implications for reducing energy dependence ..... 17

***6. Electrification of Final Uses*** ..... 18

6.1. A cost–benefit illustration: gas boiler vs. heat pump in the Italian case ..... 18

6.2. Economic Analysis ..... 19

6.3. Application to Italy ..... 19

6.4. Main Limits and Trade-Offs ..... 20

6.5. Implications for Reducing Energy Dependence ..... 21

6.6. Comparison Table ..... 21

***7. Renewable Electricity: Solar and Wind as Domestic Supply*** ..... 23

7.1. Brief introduction ..... 23

7.2. Economic analysis ..... 23

7.3. Application to Italy ..... 23

7.4. Main limits and trade-offs ..... 24

7.5. Implications for reducing energy dependence ..... 24

7.6. Comparison table ..... 25

***8. Grid, Storage, Interconnections and System Flexibility*** ..... 27

8.1. Brief introduction ..... 27

8.2. Economic analysis ..... 27

8.3. Application to Italy ..... 28

8.4. Main limits and trade-offs ..... 28

8.5. Implications for reducing energy dependence ..... 29
8.6. Comparison table ..... 29
***9. Biomethane, Hydrogen and Hard-to-Electrify Sectors*** ..... 31
9.1. Brief introduction ..... 31
9.2. Economic analysis ..... 31
9.3. Application to Italy ..... 32
9.4. Main limits and trade-offs ..... 32
9.5. Implications for reducing energy dependence ..... 32
9.6. Comparison table ..... 33
***10. Final Comparative Assessment: Which Options Reduce Italy's Energy Dependence the Most?*** ..... 35
10.1. Comparative matrix ..... 35
10.2. Direct, enabling and targeted measures ..... 36
10.3. Ranking of the options ..... 37
10.4. Short-term versus long-term contribution ..... 37
10.5. Main trade-offs ..... 38
10.6. Overall result ..... 38
***11. Policy Implications: What Italy Should Do*** ..... 40
11.1. Put energy efficiency first ..... 40
11.2. Accelerate renewable deployment, but link it to system value ..... 40
11.3. Electrify heating and transport with correct price signals ..... 41
11.4. Treat grids, storage and demand response as strategic infrastructure ..... 41
11.5. Use biomethane and hydrogen selectively ..... 42
11.6. Protect affordability and industrial competitiveness ..... 42
11.7. Improve governance and coordination ..... 42
11.8. Overall policy package ..... 43
***12. Conclusions*** ..... 44
***Bibliography*** ..... 46

# *1. Introduction*

Energy dependence represents one of the main structural vulnerabilities of the Italian economy. Italy relies heavily on imported energy sources, especially natural gas and oil, which exposes the country to international price volatility, geopolitical tensions and uncertainty in energy supply. This dependence affects not only the energy sector, but also the broader economy: energy prices influence household expenditure, firms' production costs, inflation, public finances and industrial competitiveness.

The scale of the problem is significant. Italy's energy import dependency remained around 74.6% in 2023, despite a decline from the previous year. The structure of final energy demand also shows that dependence is concentrated in specific sectors: transport is still overwhelmingly based on petroleum products, while residential energy use remains strongly linked to natural gas. In the electricity sector, renewable generation is expanding, but gas-fired generation and cross-border electricity flows still play an important role. This means that Italy's energy dependence is not a generic problem of "importing too much energy", but a sector-specific vulnerability rooted in transport, heating, electricity generation and infrastructure.

The recent European energy crisis showed that energy dependence is not a purely technical issue. When international gas prices increased and supply routes became less stable, the consequences were transmitted rapidly to electricity prices, consumer bills and the cost structure of energy-intensive industries. For a country such as Italy, where domestic fossil fuel resources are limited, reducing energy dependence is therefore both an economic priority and a strategic objective.

However, the energy transition changes the meaning of energy security. In the past, energy security was often associated with securing access to fossil fuel supplies. Today, it increasingly depends on the ability to reduce fossil fuel consumption, expand domestic renewable energy production, improve energy efficiency and strengthen the flexibility of the electricity system. Reducing dependence does not simply mean replacing one supplier with another. It means redesigning the energy system so that the economy becomes less vulnerable to external shocks.

The central research question of this project is the following:

**How can Italy effectively reduce its energy dependence while balancing energy security, environmental sustainability and economic affordability?**

The main argument of the paper is that Italy's energy dependence cannot be reduced through a single technological solution. A credible strategy requires a coordinated portfolio of measures acting on both the demand and supply sides of the energy system. Energy efficiency reduces final energy demand; renewable energy increases domestic low-carbon supply; electrification replaces imported fossil fuels in heating and transport; and grid flexibility, storage and interconnections make a more renewable-based electricity system reliable and scalable. Biomethane and hydrogen can also contribute, but mainly in specific sectors where direct electrification is technically or economically difficult.

The contribution of this paper is to connect energy-security theory with the sectoral structure of the Italian energy system. The analysis shows that different sources of dependence require different policy instruments. Oil dependence in transport cannot be addressed in the same way as gas dependence in buildings or flexibility constraints in the electricity system. For this reason, the paper evaluates each option according to its contribution to import reduction, affordability, environmental sustainability, scalability, implementation time and regulatory feasibility.

The objective is not simply to describe different technologies, but to assess which combination of policies and investments can most effectively reduce Italy's energy vulnerability. The analysis therefore connects theoretical concepts from energy and environmental economics with the practical challenges of the Italian energy system.

# *2. Theoretical Framework: Energy Security, Energy Accounting and the Energy Trilemma*

The starting point of this project is the concept of energy security. Traditionally, energy security has often been interpreted as the ability of a country to guarantee sufficient and reliable energy supplies. In this narrow view, the main policy concern is the physical availability of fuels, especially oil and natural gas. However, this definition is not sufficient for the analysis of a modern energy system. Energy security is not only about having access to energy resources, but also about the economic, technological, environmental and institutional conditions that make energy services reliable, affordable and sustainable.

This broader interpretation is particularly relevant for Italy. Italy is not only exposed to the risk of insufficient energy supply; it is exposed to a combination of vulnerabilities: high dependence on imported fossil fuels, sensitivity to international price volatility, limited domestic fossil resources, infrastructure constraints and the need to decarbonise the energy system. For this reason, reducing energy dependence cannot mean achieving complete self-sufficiency. Instead, it should be understood as reducing systemic vulnerability.

A useful framework is provided by Sovacool and Mukherjee, who conceptualise energy security as a multidimensional phenomenon. Their approach identifies five main dimensions: availability, affordability, technology development and efficiency, environmental and social sustainability, and regulation and governance. This framework is useful because it avoids reducing energy security to a single indicator, such as import dependence, and instead highlights the interactions between supply, demand, prices, infrastructure, environmental constraints and policy design.

The first dimension is **availability**. Availability refers to the capacity of a country to access sufficient energy resources and energy services. In the Italian case, availability is closely connected to dependence on imported gas and oil. A country that imports a large share of its energy is vulnerable to disruptions in international markets, geopolitical tensions and changes in the strategies of exporting countries. However, availability can be improved in several ways: by increasing domestic renewable energy production, by diversifying sources and suppliers, by reducing final energy demand and by strengthening infrastructure.

The second dimension is **affordability**. Energy security also requires energy services to be supplied at stable and reasonable prices. High energy prices affect households through higher bills and transport costs, firms through higher production costs, and governments through the need for subsidies or emergency measures. For Italy, import dependence becomes an economic problem because international fossil fuel price volatility can be transmitted to domestic inflation, industrial competitiveness and public finances. Therefore, a strategy to reduce energy dependence must also consider the cost of the transition and the distribution of those costs across consumers, firms and the public sector.

The third dimension is **technology development and efficiency**. A secure energy system must be able to adapt to disruptions, invest in innovation and improve the efficiency with which energy is used. Energy efficiency is central because the cheapest and most secure unit of energy is often the one that is not consumed. Lower final energy demand reduces the need for imported fuels, lowers exposure to price shocks and decreases pressure on infrastructure. At the same time, technologies such as heat pumps, electric vehicles, batteries, smart grids and demand response can reduce fossil fuel consumption and increase system resilience. However, these technologies require investment, regulation and coordination.

The fourth dimension is **environmental and social sustainability**. Energy dependence cannot be reduced by simply replacing imported fossil fuels with domestic high-carbon sources. Such a strategy would conflict with climate targets and would increase environmental externalities. The energy transition therefore changes the meaning of energy security: a secure energy system must also be compatible with decarbonisation. Renewable energy, energy efficiency and electrification are relevant because they can simultaneously reduce imported fossil fuel demand and lower greenhouse gas emissions.

The fifth dimension is **regulation and governance**. Energy systems are strongly shaped by public policy, market rules and institutional coordination. Energy markets are affected by several market failures, including environmental externalities, information asymmetries, network externalities and underinvestment in public goods such as security of supply and grid resilience. For this reason, policy instruments such as carbon pricing, renewable energy auctions, efficiency standards, grid regulation, capacity mechanisms and incentives for storage and flexibility are essential. Without effective governance, even technically promising solutions may fail to scale.

A second theoretical element is **energy accounting**. Import dependence cannot be understood only by looking at total energy imports. It must be analysed through the energy balance, which distinguishes between primary energy supply, transformation processes, final energy consumption and electricity generation. This distinction is crucial because different policy options act on different parts of the energy system. Energy efficiency reduces final energy demand; renewables increase domestic primary and secondary energy supply; electrification shifts final consumption from fossil fuels to electricity; and flexibility improves the ability of the electricity system to integrate variable renewable generation.

This distinction also prevents misleading conclusions. A country may reduce direct gas use in buildings by electrifying heating, but if the additional electricity is generated by gas-fired power plants or imported electricity, the reduction in energy dependence is weaker. Similarly, expanding renewable capacity reduces fossil fuel dependence only if the grid can absorb and deliver that electricity to final consumers. Energy accounting therefore provides the empirical foundation for evaluating whether a policy truly reduces dependence or merely shifts it from one part of the system to another.

A third element is the **energy trilemma**, which describes the need to balance three objectives: energy security, affordability and environmental sustainability. These objectives can reinforce each other, but they can also create trade-offs. Renewable energy can improve sustainability and reduce fossil fuel imports, but it requires grid investment and flexibility. Electrification can reduce gas and oil consumption, but it increases electricity demand and may create pressure on the power system. Energy efficiency can reduce demand and costs, but it may face behavioural barriers, financial constraints and rebound effects.

The theoretical implication is that energy dependence is not only a matter of import volumes. It is a systemic issue involving the structure of demand, the composition of supply, the organisation of energy markets, infrastructure investment and public regulation. Therefore, this project evaluates each option not only according to its direct effect on imports, but also according to its impact on prices, emissions, scalability, reliability and policy feasibility.

In this perspective, reducing Italy’s energy dependence requires a coordinated portfolio of measures. Energy efficiency acts on demand; renewables act on domestic supply; electrification changes the energy carriers used in final consumption; grid flexibility enables system integration; and biomethane and hydrogen provide targeted solutions for sectors where direct electrification is difficult. The role of policy is to combine these instruments in a way that reduces vulnerability without undermining affordability, competitiveness or environmental objectives.

# *3. The Italian Case: The Current Structure of the Energy System*

Italy's energy dependence is not simply a matter of "importing too much energy". It is the result of a specific structural configuration: limited domestic fossil fuel resources, a final energy demand still strongly based on oil and natural gas, and an electricity system that is becoming cleaner but still relies on gas-fired generation and cross-border exchanges. For this reason, understanding the Italian case requires an energy-accounting approach. The same country can appear more or less dependent depending on whether the analysis focuses on primary energy supply, final energy consumption, or electricity generation.

This distinction is essential for the research question of the paper. Reducing energy dependence does not only mean increasing domestic production. It also means reducing the amount of imported fuels needed to provide the same energy services, changing the energy carriers used in final consumption, and making the electricity system able to integrate larger shares of domestic renewable generation. In this sense, the Italian case offers a clear example of the energy trilemma: policies must improve security of supply without compromising affordability and environmental sustainability.

## 3.1 Import Dependence and Primary Energy Supply

Italy remains one of the most energy-dependent economies in Europe. According to the latest national energy balance published by MASE, Italy's import dependency ratio decreased from 79.2% in 2022 to 74.6% in 2023. This decline is relevant because it indicates some improvement in domestic production and a reduction in net energy imports. However, the level remains extremely high. In practical terms, most of the energy used by Italian households, firms and transport systems is still directly or indirectly linked to international energy markets.

This dependence is mainly concentrated in fossil fuels. Oil products dominate transport, while natural gas plays a central role in residential heating, industrial processes and electricity generation. The problem is therefore not only the quantity of imports, but also their macroeconomic relevance. When international oil or gas prices increase, the shock is transmitted to fuel prices, electricity prices, firms' production costs, household expenditure and inflation. The recent European energy crisis showed that import dependence is not a purely technical vulnerability: it can rapidly become a competitiveness problem for industry and a social problem for households.

From the perspective of energy security, this confirms the multidimensional framework introduced in Chapter 2. Availability matters because Italy must secure sufficient energy supplies from abroad. Affordability matters because imported fossil fuels expose the country to volatile international prices. Sustainability matters because reducing import dependence by increasing domestic fossil fuel use would be inconsistent with climate objectives. Regulation and governance matter because the transition requires coordinated investment in networks, renewables, efficiency and flexibility.

## 3.2 Final Energy Consumption: Where Energy Is Actually Used

A second important point is that energy dependence must be studied through final energy consumption, not only through primary energy supply. Final consumption shows where energy is actually used by households, firms and transport users. This is crucial because the most effective policies are sector-specific: the solution for transport is not the same as the solution for residential heating or for energy-intensive industry.

In 2022, Italy's final energy consumption was approximately 109.3 Mtoe. The composition by energy carrier shows that fossil fuels still dominated final demand. Oil and petroleum products represented the largest component, followed by natural gas and electricity. Renewable energy and biofuels played an increasing but still smaller role.

**Table 3.1. Approximate structure of final energy consumption in Italy, 2022**

| Energy carrier | Main role in the Italian system | Relevance for energy dependence |
|---|---|---|
| Oil and petroleum products | Mainly transport, agriculture and some industrial uses | Very high, because crude oil and refined products are largely imported |
| Natural gas | Residential heating, industry, services and power generation | Very high, because Italy has limited domestic gas production |
| Electricity | Industry, services, households and growing electrification | Medium to high, depending on how electricity is generated |
| Renewables and biofuels | Heating, bioenergy, renewable electricity and transport biofuels | Positive for reducing dependence, but still limited in scale |
| Derived heat and other sources | District heating and minor uses | Limited but relevant in specific local systems |

Source: Authors' elaboration based on MASE energy balance data.

This structure has an important implication: Italy's dependence cannot be reduced only by increasing renewable electricity. Electricity is only one part of final energy consumption. A large share of Italian energy use still takes place directly through fuels: diesel and gasoline in transport, methane in buildings, and gas in industrial processes. Therefore, the transition must act simultaneously on three channels: reducing final demand through efficiency, replacing fossil fuels with electricity where possible, and producing that electricity increasingly from domestic low-carbon sources.

## 3.3 Sectoral Exposure: Transport, Buildings and Industry

The sectoral structure of final energy demand helps identify the main vulnerabilities. Transport, residential buildings and industry are the three most important areas for understanding Italy's energy dependence.

Transport is the most oil-dependent sector. Road transport relies heavily on diesel and gasoline, which makes it directly exposed to international oil markets. This creates a strong link between energy dependence and the balance of payments, because transport fuel demand requires large imports of crude oil or refined products. It also makes households and firms vulnerable to fuel price increases. For this reason, electrification of mobility, modal shift, public transport and efficiency improvements in vehicles are not only environmental policies, but also energy security policies.

The residential sector is mainly vulnerable through natural gas. Heating demand is a central component of household energy use, and Italy has historically relied heavily on gas boilers. This creates a seasonal vulnerability: gas demand rises in winter, exactly when supply security and price volatility can become more critical. Heat pumps, building renovation and thermal insulation can therefore reduce import dependence by lowering the amount of gas required to provide heating services.

Industry is more heterogeneous. Some industrial sectors use electricity intensively, while others depend on natural gas for heat and processes. Energy-intensive industries are particularly exposed because energy prices affect their competitiveness. For these sectors, energy dependence is not only an issue of supply security, but also of production costs. Efficiency, electrification of low- and medium-temperature heat, renewable power

purchase agreements and targeted use of biomethane or hydrogen in hard-to-electrify processes can all contribute, but the feasibility differs across subsectors.

**Table 3.2. Main sectoral vulnerabilities in the Italian energy system**

| Sector | Main energy dependence channel | Main policy implication |
|---|---|---|
| Transport | Oil products | Electric vehicles, public transport, modal shift, biofuels for specific uses |
| Residential buildings | Natural gas for heating | Heat pumps, insulation, efficiency standards, renovation incentives |
| Industry | Gas and electricity costs | Process efficiency, electrification, renewable PPAs, targeted low-carbon gases |
| Services | Electricity and gas | Efficiency, heat pumps, distributed solar, demand response |
| Agriculture | Diesel and electricity | Efficiency, electrification of machinery where feasible, biogas/biomethane |

This sectoral reading is important because it prevents an overly simplistic conclusion. Italy is not dependent on a single energy source for a single use. It is dependent through multiple channels: oil for mobility, gas for heating and industry, gas for electricity generation, and electricity imports during specific periods. A credible strategy must therefore be differentiated by sector.

## 3.4 The Electricity System: Cleaner, but Still Strategically Central

The electricity system is the core infrastructure of the energy transition. Electrification of transport, heating and some industrial processes can reduce direct consumption of oil and gas. However, this strategy works only if the electricity system becomes cleaner, more reliable and less dependent on imported fuels.

In 2024, Italy's electricity demand was around 311.9 TWh. Most demand was met by domestic production, but net electricity imports still covered a significant share. Terna reports imports of 55.9 TWh and exports of 4.9 TWh in 2024, implying net imports of about 51.0 TWh. This does not necessarily represent a weakness in itself: interconnections can improve security by allowing countries to exchange electricity and balance supply and demand. However, it does show that Italy's electricity security also depends on the European electricity system.

The domestic electricity mix is improving. According to GSE, in 2024 renewable sources accounted for 51.83% of the national electricity production injected into the system, while natural gas accounted for 42.01%. Coal and petroleum products played a much smaller role. This confirms that the power sector is moving in the right direction, but it also highlights a key vulnerability: gas remains central in electricity generation and often influences wholesale electricity prices.

This point is essential for the rest of the project. If Italy electrifies transport and heating while electricity is still partly produced from imported gas, the country reduces direct oil and gas consumption in final uses, but it may increase pressure on the power system. Therefore, electrification and renewable expansion must proceed

together. A larger renewable electricity system also requires more flexibility, storage, demand response, grid investment and interconnections.

The economics of electricity reinforce this idea. Electricity is not only energy measured in kWh. A reliable power system must also provide capacity, flexibility, frequency control, voltage stability and adequacy at specific times and locations. This is why renewable expansion cannot be evaluated only through average generation costs. Solar and wind reduce fuel imports and emissions, but their value depends on when and where they produce, how much flexibility is available, and whether the grid can absorb their output.

## 3.5 Main Structural Criticalities of the Italian System

The Italian energy system presents five major criticalities.

First, fossil fuel import dependence remains high. Even after the decline observed in 2023, Italy still imports most of the energy it uses. This makes the economy exposed to geopolitical risk, exchange rate movements, international price volatility and supply disruptions.

Second, final demand is still fossil-intensive. The problem is not only how electricity is generated, but also how energy is consumed. Transport is still oil-based, buildings are still gas-based, and parts of industry remain difficult to electrify.

Third, gas has a double role. It is used directly in households, services and industry, but also indirectly through electricity generation. This creates a strong link between gas markets and electricity prices. As a result, a gas price shock can affect both heating bills and electricity bills.

Fourth, renewable growth creates an integration challenge. More solar and wind reduce fuel imports and emissions, but they increase the need for grid capacity, storage and flexible demand. Without these complementary investments, renewable electricity may face curtailment, congestion and lower system value.

Fifth, policy coordination is essential. Energy dependence cannot be reduced by isolated measures. Incentives for heat pumps are less effective without building efficiency. Renewable auctions are less effective without grid connection. Electric vehicles are less beneficial without charging infrastructure and smart charging. Biomethane and hydrogen are useful only if targeted to sectors where direct electrification is not feasible.

## 3.6 What the Italian Case Implies for the Research Question

The diagnosis of the Italian energy system leads to a clear conclusion: Italy’s energy dependence is structural, sectoral and systemic. It cannot be solved by one technology alone.

Renewables are necessary because they increase domestic low-carbon supply and reduce the need for imported fuels in electricity generation. However, they are not sufficient without grid flexibility, storage and market rules that reward system value.

Energy efficiency is essential because it reduces the amount of energy required to provide the same services. In the Italian case, this is particularly important for buildings and industry, where lower demand directly reduces gas consumption and price exposure.

Electrification is a powerful strategy because it can replace imported oil and gas in transport and heating. However, it increases electricity demand and therefore requires a cleaner and more resilient power system.

Flexibility is not an optional complement, but an enabling condition. Storage, demand response, smart grids and interconnections determine whether high shares of renewables can be integrated reliably and economically.

Finally, biomethane and hydrogen should be understood as targeted solutions, not as universal substitutes. Their role is most credible in hard-to-electrify sectors, where direct electrification is technically difficult or too costly.

Therefore, the Italian case confirms the main argument of this paper: reducing energy dependence requires a coordinated portfolio of measures. The objective is not full energy autarky, which would be unrealistic and inefficient, but lower systemic vulnerability. Italy can become less dependent by consuming less energy, using more domestic renewable electricity, electrifying final uses where efficient, and strengthening the flexibility and governance of the energy system.

# *4. Methodology: A Multi-Criteria Assessment of Energy Dependence Reduction Options*

This chapter explains the methodology used to compare the main strategies available to Italy for reducing energy dependence. The project adopts a qualitative multi-criteria assessment rather than a quantitative energy-system model. This choice is appropriate because the options analysed in the paper are different in nature: some reduce energy demand, some increase domestic supply, some substitute fossil fuels in final uses, while others enable the electricity system to integrate higher shares of renewables. A single numerical indicator would therefore be too narrow to capture their full economic and policy relevance.

The methodology builds on the diagnosis developed in Chapter 3. Italy's dependence is not concentrated in one single part of the energy system. Oil dependence is mainly connected to transport; gas dependence is linked to residential heating, industry and electricity generation; and electricity dependence is shaped by the interaction between domestic generation, imports, grids and flexibility. For this reason, each solution must be evaluated according to the specific vulnerability it addresses. A policy that reduces oil consumption in transport does not solve gas dependence in buildings, and a policy that expands renewable electricity does not automatically solve grid constraints.

The first analytical foundation is energy accounting. Energy dependence is assessed by distinguishing between primary energy supply, final energy consumption and electricity generation. This distinction is essential because different policy options operate at different stages of the energy system. Energy efficiency reduces final energy demand; renewable energy increases domestic low-carbon supply; electrification changes the energy carriers used in final consumption; and flexibility determines whether renewable electricity can be effectively integrated into the system. Without this accounting logic, there is a risk of confusing a real reduction in dependence with a simple shift of dependence from one sector to another.

The second analytical foundation is the multidimensional concept of energy security. Following Sovacool and Mukherjee, energy security is interpreted not only as physical availability of supply, but also as affordability, technological resilience, environmental sustainability and regulatory quality. This is important because a measure may reduce imports while generating other trade-offs. For example, electrification can reduce direct gas and oil consumption, but it can also increase electricity demand and require large grid investments. Similarly, renewables reduce fossil fuel use and emissions, but their system value depends on location, timing, storage, interconnections and market design.

The analysis compares five categories of solutions:

1. energy efficiency;
2. electrification of final uses, especially heat pumps and electric vehicles;
3. renewable electricity, especially solar and wind;
4. grid flexibility, storage and interconnections;
5. biomethane and hydrogen for hard-to-electrify sectors.

Each option is evaluated according to eight criteria.

| Criterion | Meaning in the analysis |
|---|---|
| Import reduction | The extent to which the option reduces Italy's need for imported gas, oil, electricity or other energy carriers. |

| | |
|---|---|
| Energy security | The contribution to reducing exposure to geopolitical risks, supply disruptions and external market shocks. |
| Affordability | The expected impact on households, firms, public finances and energy prices. |
| Implementation time | Whether the option can produce effects in the short, medium or long term. |
| Scalability | The potential for large-scale deployment in the Italian context. |
| Environmental impact | The contribution to reducing greenhouse gas emissions and other environmental externalities. |
| System integration | The need for complementary infrastructure such as grids, storage, demand response or market redesign. |
| Regulatory feasibility | The degree to which the option depends on permits, incentives, market rules, public planning and institutional coordination. |

The criteria are applied through a qualitative ranking based on three levels: high, medium and low. A “high” score means that the option can make a large and direct contribution to reducing dependence, or that it is essential for enabling other measures. A “medium” score means that the contribution is relevant but conditional on context, scale, infrastructure or policy design. A “low” score means that the contribution is limited, indirect, long-term or restricted to specific sectors.

The assessment distinguishes between direct measures and enabling measures. Energy efficiency and renewables are direct measures. Efficiency reduces the amount of energy needed to provide heating, mobility, industrial output and other energy services. Renewable electricity increases domestic low-carbon production and reduces the need for fossil fuels in power generation. Electrification is also a direct substitution measure, but its effectiveness depends on the electricity mix. If additional electricity demand is met by domestic renewables, electrification reduces dependence substantially. If it is met by gas-fired generation or imported electricity, the reduction is weaker.

Grid flexibility, storage and interconnections are enabling measures. They do not reduce fossil fuel imports in the same immediate way as efficiency or renewables, but they determine whether a more renewable and electrified system can function reliably. A high-renewables electricity system requires balancing capacity, demand response, storage, stronger transmission and distribution networks, and market rules that reward flexibility. Without these elements, renewable expansion may lead to congestion, curtailment, reliability problems and higher balancing costs.

Biomethane and hydrogen are treated as targeted measures. Their role is assessed mainly in relation to sectors where direct electrification is difficult or expensive, such as some industrial processes, heavy transport and specific thermal uses. They are not evaluated as universal substitutes for natural gas or oil, because their availability, production costs and infrastructure requirements limit their scalability.

The methodology also includes a time dimension. Some options can reduce dependence relatively quickly, while others require long investment cycles. Energy efficiency can produce benefits in the short to medium term, especially in buildings and industry, although financial barriers and behavioural constraints may slow

adoption. Renewable deployment can generate a strong structural reduction in fossil fuel dependence, but it depends on permitting, grid capacity and investment conditions. Electrification has high long-term potential, but it requires the replacement of durable assets such as boilers, vehicles and industrial equipment. Grid investment, storage and hydrogen infrastructure are also long-term processes, even if they are necessary for the credibility of the transition.

This approach has two main advantages. First, it avoids treating all technologies as substitutes. The options analysed in this paper are mostly complementary: efficiency reduces the scale of the problem, electrification changes the form of final demand, renewables provide domestic low-carbon electricity, and flexibility allows the system to integrate variable generation. Second, it connects technology assessment with policy design. The same technology can have very different effects depending on incentives, regulation, tariffs, permitting procedures and market rules.

The final output of the methodology is a comparative matrix presented in Chapter 10. The matrix does not claim to provide exact numerical estimates of avoided imports or costs. Instead, it offers a structured qualitative comparison that identifies which measures are most effective, which are complementary, which are constrained by implementation barriers, and which should be reserved for specific sectors.

This methodological choice is consistent with the logic of the energy trilemma. A policy that improves security but ignores affordability may generate social and political resistance. A policy that reduces emissions but weakens reliability may undermine energy security. A policy that reduces imports but creates excessive costs may damage competitiveness. Therefore, the purpose of the comparison is not simply to minimize imports, but to identify a balanced portfolio capable of reducing Italy's vulnerability while supporting sustainability, affordability and economic resilience.

# *5. Energy Efficiency: Reducing Demand Before Replacing Supply*

Energy efficiency is the first demand-side instrument in a credible strategy to reduce Italy's energy dependence. While renewable energy increases domestic low-carbon supply and electrification changes the energy carrier used in final consumption, energy efficiency acts before both of them: it reduces the amount of energy required to obtain the same energy service. This distinction is central. Households and firms do not demand energy for its own sake; they demand heating, cooling, lighting, mobility, mechanical power and industrial output. Energy is therefore an input into the production of energy services, and efficiency improves the ratio between useful service and energy input.

This makes energy efficiency different from simple energy conservation. Conservation means reducing total energy consumption, possibly by consuming fewer energy services. Efficiency, instead, allows the same or a similar level of service to be delivered with less energy. For Italy, this distinction is particularly important because the objective is not to reduce welfare or industrial activity, but to lower the fossil-fuel intensity of households, firms and transport. In this sense, efficiency is a security policy, an environmental policy and an affordability policy at the same time.

## 5.1. Economic rationale

From an economic perspective, energy efficiency can be interpreted as an investment decision. Consumers and firms often face a trade-off between a higher upfront capital cost and lower future operating costs. Better insulation, efficient boilers, heat recovery systems, efficient motors, LED lighting or advanced industrial controls usually require an initial investment, while the benefits arrive gradually through lower energy bills. The decision therefore depends on expected energy prices, intensity of use, equipment lifetime, discount rates, access to finance and uncertainty about future savings.

In a perfectly functioning market, households and firms would choose the level of efficiency that minimises the present value of total costs. However, actual energy-efficiency choices often differ from this benchmark. This is the well-known "energy efficiency gap": many investments that appear cost-effective from a technical or social perspective are not adopted, or are adopted too slowly. The gap does not mean that all possible efficiency improvements should be implemented regardless of cost. Rather, it means that several market and behavioural failures may prevent privately optimal decisions from aligning with socially desirable outcomes.

The main failures are information problems, split incentives, capital constraints, behavioural biases and environmental externalities. Information problems arise because consumers may not know the future savings associated with efficient technologies. Split incentives occur when the person making the investment is not the person paying the energy bill, as in rented housing where landlords decide the efficiency level but tenants pay for heating and electricity. Capital constraints matter because many households and small firms cannot finance investments with long payback periods. Behavioural biases, such as myopia or excessive focus on upfront costs, can further reduce adoption. Finally, environmental and security externalities mean that private energy prices do not fully reflect the social cost of fossil-fuel use, including emissions and import vulnerability.

For these reasons, energy efficiency is not only a technical issue. It is an economic and institutional problem. The relevant question is not simply whether efficient technologies exist, but whether prices, incentives, information and regulation allow them to diffuse at the scale required.

## 5.2. Why energy efficiency matters for Italy

Energy efficiency is especially relevant in the Italian case because Italy's energy dependence is concentrated in specific sectors and fuels. As shown in Chapter 3, final energy demand is still dominated by fossil fuels, with petroleum products central in transport and natural gas central in residential heating and industry. This means that reducing energy dependence cannot be achieved only by adding domestic renewable electricity. Italy must also reduce the amount of gas, oil and electricity required to provide final energy services.

Buildings are the first priority. The residential sector is structurally exposed to natural gas because a large share of Italian households still relies on gas boilers for space heating and domestic hot water. Improving building envelopes through insulation, better windows, efficient heating systems and smart controls reduces gas demand directly. This lowers exposure to imported gas, reduces household bills and makes future electrification through heat pumps easier, because a well-insulated building requires a smaller heating system and less electricity during winter peaks.

Industry is the second major area. Industrial energy efficiency includes waste-heat recovery, process optimisation, efficient motors, variable-speed drives, cogeneration where appropriate, digital energy management and fuel-saving technologies. These measures reduce both gas and electricity demand. In energy-intensive sectors, efficiency also improves competitiveness because energy prices are a key component of production costs. This is crucial for Italy, where energy-price shocks can rapidly affect firms' margins, export competitiveness and inflation.

Transport is different. In this sector, efficiency improvements in internal-combustion vehicles can reduce oil demand, but they cannot eliminate dependence on petroleum products. This is why efficiency in transport should be interpreted as complementary to modal shift and electrification. More efficient vehicles, better logistics, public transport, rail freight and urban planning can reduce energy demand, while electric mobility changes the energy carrier. Efficiency therefore prepares the ground for the deeper transformation discussed in the electrification chapter.

## 5.3. Policy instruments and the "failure-matching" principle

Because energy efficiency is affected by multiple failures, policy design must match the specific barrier it wants to correct. A single instrument cannot solve all problems. If the problem is lack of information, the appropriate tools are energy labels, audits, information campaigns and transparent performance certificates. If the problem is split incentives, minimum standards for rented buildings or renovation obligations may be more effective than generic subsidies. If the problem is liquidity, concessional loans, guarantees or targeted grants can help households and small firms overcome upfront costs. If the problem is an environmental externality, carbon pricing or energy taxation can improve the relative price of efficient choices.

This "failure-matching" principle is important for avoiding wasteful policy. Subsidies can accelerate adoption, but they also risk free-riding if they reward households or firms that would have invested anyway. They can also inflate prices if supply capacity in the construction sector is limited. The Italian Superbonus experience is useful in this respect. It demonstrated that generous incentives can stimulate renovation activity, but also that policy design must preserve cost discipline, target additional investment and avoid excessive fiscal burdens. The lesson is not that building renovation is unimportant; the lesson is that the instrument must be designed carefully.

European policy is moving in this direction. The revised Energy Efficiency Directive strengthens the role of efficiency in the EU decarbonisation strategy, while the revised Energy Performance of Buildings Directive focuses on increasing renovation rates and improving the worst-performing buildings. These policies are directly relevant to Italy because the building stock is long-lived, heterogeneous and often inefficient. Without regulatory and financial support, renovation would proceed too slowly to make a meaningful contribution to energy security by 2030.

## 5.4. Limits and trade-offs

Energy efficiency is often described as the cheapest energy resource, but it is not unlimited and it is not automatic. Three limits are particularly important.

The first is the rebound effect. When efficiency lowers the cost of an energy service, consumers may use more of that service. For example, after insulation, households may heat more rooms or keep indoor temperatures higher. After purchasing a more efficient car, drivers may travel more because each kilometre is cheaper. This means that technical energy savings are usually larger than realised energy savings. However, rebound should

not be interpreted only negatively. Part of it reflects a welfare gain, especially for households that were previously under-consuming heating or cooling. The policy implication is that efficiency should be combined with appropriate energy pricing and carbon pricing, not abandoned.

The second limit is timing. Buildings, vehicles and industrial equipment are long-lived capital stocks. Efficiency improvements therefore depend on renovation cycles, replacement rates and investment capacity. Even when a technology is cost-effective, diffusion may take years. This matters for energy security because efficiency can reduce vulnerability structurally, but it cannot solve a sudden supply crisis overnight.

The third limit is distributional. Upfront costs can exclude low-income households, renters and small firms. If policies rely only on tax deductions, they may mainly benefit households with sufficient taxable income and access to credit. A socially credible efficiency strategy must therefore include targeted finance, support for vulnerable households and mechanisms that allow tenants and small firms to participate.

## 5.5. Implications for reducing energy dependence

Energy efficiency should be evaluated as a foundational measure in Italy's energy-dependence strategy. It reduces imported gas in buildings and industry, lowers oil demand where transport efficiency and modal shift are possible, and reduces electricity demand during peak periods. It also makes other strategies cheaper. Every unit of demand avoided is a unit that does not need to be electrified, generated by renewables, transported through the grid or balanced through storage.

Using the criteria defined in Chapter 4, energy efficiency performs strongly on import reduction, affordability and environmental sustainability. Its investment needs are significant but distributed across households, firms and public buildings. Its main weakness is not technical potential, but implementation: slow renovation rates, financing barriers, split incentives and rebound effects limit the speed and scale of realised savings. For this reason, energy efficiency is best understood as a "low-regret" policy, but not as a stand-alone solution.

The key conclusion is that the most secure unit of energy is the one that Italy does not need to import, purchase, transport or burn. Efficiency reduces the size of the energy-dependence problem before supply-side solutions are added. In the overall portfolio of this paper, it should therefore come first: efficiency shrinks final demand, electrification changes the remaining fuel base, renewables supply an increasing share of domestic electricity, and grid flexibility makes the new system reliable.

# *6. Electrification of Final Uses*

Efficiency reduces demand for a given fuel; electrification changes the fuel itself. By moving heating and mobility from gas and oil to electricity, end uses that today rely heavily on imported fossil fuels can, in principle, be supplied by domestically generated electricity. This makes electrification one of the most important strategies for reducing the two main pillars of Italian energy dependence: imported gas for buildings and imported oil for transport.

However, electrification also creates a central trade-off. It does not eliminate energy demand; it transfers part of that demand to the power system. Therefore, its contribution to energy security depends on how electricity is generated, when it is consumed, and whether the grid can manage the additional load. If new electric demand is met mainly by domestic renewable generation, electrification reduces dependence. If it is met by gas-fired generation or imported electricity, the reduction is weaker and the dependence may simply move from final consumption to the electricity system.

In demand-management terms, electrification is a substitution between energy carriers. It is driven by technology, relative prices and policy. It also connects the demand side of the energy system to the supply and market side: the value of an electrified kilowatt-hour depends on the marginal power plant that serves it, on the timing of consumption and on the availability of networks, storage and flexibility. This is why electrification cannot be evaluated as a stand-alone measure. It must be analysed together with renewables, grid investment and market design.

## 6.1. A cost–benefit illustration: gas boiler vs. heat pump in the Italian case

To make the argument of this chapter concrete, we apply the cost–benefit logic to the replacement of a condensing gas boiler with an air-source heat pump in a typical Italian dwelling. We follow the standard steps of cost–benefit analysis: we define the gas boiler as the base policy, set a time horizon equal to the equipment lifetime (about 15–20 years), monetise costs and benefits, and discount future flows to present value.

The dwelling requires about 13 MWh of useful heat per year, covered either by roughly 1,400 m³ of natural gas (condensing boiler, seasonal efficiency ≈ 0.90) or by a heat pump with a seasonal coefficient of performance of about 3, which needs around 4,300 kWh of electricity. Using average cost pricing — the cost per unit of useful service — and the all-in retail prices observed in Italy in 2025 (electricity ≈ 0.33 €/kWh; gas ≈ 1.05 €/m³, i.e. ≈ 0.10 €/kWh):

ItemGas boilerHeat pump (COP 3) Energy input per year≈ 1,400 m³ gas≈ 4,300 kWh electricityAll-in price≈ 0.10 €/kWh≈ 0.33 €/kWhAnnual operating cost≈ 1,470 €≈ 1,430 €Cost per kWh of useful heat≈ 0.113 €≈ 0.110 €Direct $CO_2$ emissions per year≈ 2.7 $tCO_2$≈ 1.1 $tCO_2$ (grid)

The result is instructive. The two options have almost identical cost per unit of useful heat, so the annual operating saving is close to zero (≈ 40 €). The break-even point falls at a COP of about 3 — precisely the level a real air-source heat pump reaches in the northern Italian climate. In other words, at current Italian price ratios the heat pump barely pays for its own electricity, even though it is physically far more efficient.

The reason lies in the way electricity is priced. In the wholesale market the price is set by the marginal generating unit, which in Italy is frequently a gas-fired plant; the retail price is then further raised by network charges and taxes embedded in the two-part tariff structure. As a consequence, electricity costs roughly three times as much as gas per kWh, and this ratio almost exactly offsets the efficiency advantage of the heat pump. This is the same conditionality stressed throughout the chapter: electrification reduces dependence only when the marginal electricity is no longer produced from imported gas.

Once the heat pump's higher upfront cost is included (an incremental investment of roughly 9,000 €), the private net present value of the switch is strongly negative: with a near-zero annual saving, no realistic discount rate makes the investment privately attractive. The decision is therefore driven almost entirely by public support, not by energy savings.

From a social standpoint, however, the calculation changes. The switch avoids about 1.6 $tCO_2$ per year. If this externality is internalised through a carbon price — whether a carbon tax or an emissions-trading scheme — it adds roughly 100–130 €/year of value that the household does not capture. Adding the (also un-priced) security externality of avoided gas imports widens the gap further. The wedge between the negative private net present value and the more favourable social value is exactly a market failure: an un-internalised externality combined with a distorted price signal. This is what justifies policy intervention — targeted subsidies, minimum standards or carbon pricing — rather than any claim that heat pumps are inefficient. The private signal is weak not because the technology fails, but because private prices do not reflect the social value of reduced imports and emissions.

## 6.2. Economic Analysis

Two features make electrification attractive even before considering import dependence.

First, many electric end uses are more efficient at the point of use than fossil-fuel alternatives. A heat pump does not produce heat by combustion; it moves heat from one place to another. For this reason, it can deliver roughly three units of heat for one unit of electricity, depending on climate, building characteristics and technology. Similarly, battery-electric vehicles convert a much larger share of final energy into motion than internal-combustion vehicles. Even when electricity is partly generated from fossil fuels, the total primary energy required to provide the same service often falls.

Second, electricity is the energy carrier most easily produced from domestic renewable sources. Solar and wind do not directly replace the gasoline burned in cars or the gas burned in boilers unless final demand is electrified. Electrification therefore makes demand accessible to domestic renewable supply. This is particularly important for Italy, where domestic fossil resources are limited but renewable potential is significant.

The main economic limit is on the load side. Electrification increases electricity demand and changes its profile. Heat pumps can increase winter electricity demand, especially during cold evening hours, when solar production is low. Electric vehicles can increase evening peak demand if charging is not managed. Whether this new demand is cheap or expensive depends on the marginal plant that supplies it. If the marginal unit is a gas-fired plant using imported fuel, electrified heating and transport reduce direct fossil-fuel use but may increase fossil-fuel demand in electricity generation. If the marginal unit is domestic renewable generation, the same electrification produces a genuine reduction in imported fuels.

This is the key reason why the economic value of electrification depends on timing, location and system conditions. A kilowatt-hour consumed at noon in an area with abundant solar generation is not equivalent to a kilowatt-hour consumed during an evening peak when gas plants set the price. Electrification must therefore be combined with smart charging, time-of-use tariffs, demand response, storage and grid reinforcement.

## 6.3. Application to Italy

The Italian end-use structure shows why electrification is a necessary complement to efficiency. As shown in Chapter 3, transport accounted for about one third of final energy demand in 2022 and was still overwhelmingly oil-based. This makes transport the largest direct lever for reducing imported oil. Modal shift and public transport can reduce energy demand, but electric mobility is the main technological route for replacing gasoline and diesel in passenger transport.

Buildings are the second key area. Residential energy demand in Italy remains strongly linked to natural gas, especially for space heating and domestic hot water. Heat pumps can substitute gas boilers in many residential and tertiary buildings. Their effect on energy dependence is direct: they reduce the gas burned inside buildings and replace it with electricity. The security benefit becomes stronger as electricity is increasingly generated from domestic renewables.

Italy's electricity system is now close to the point where electrification can generate substantial security benefits, but the condition is not automatic. In 2024, electricity demand was around 312 TWh, renewables covered roughly 41% of demand, and Italy still relied on a combination of non-renewable generation and net electricity imports. This means that electrification can reduce dependence only if additional electric demand is matched by more domestic renewable generation and by adequate grid flexibility. Otherwise, adding electric load may deepen pressure on the power system rather than reduce external reliance.

A useful example is the replacement of a residential gas boiler with an air-source heat pump. Consider a typical Italian dwelling heated by a condensing gas boiler consuming around 1,400 cubic metres of gas per year, corresponding roughly to 13 MWh of useful heat. If the same heat demand is supplied by a heat pump with a seasonal coefficient of performance of about 3, the dwelling would require around 4,500 kWh of electricity. The household would no longer burn gas directly, and the imported-fuel content of heating would fall as the electricity mix becomes more renewable.

However, the private economics of this switch is not automatic. The heat pump has a higher upfront cost than a gas boiler, so the decision depends on expected energy prices, access to finance and payback time. The operating advantage also depends on the ratio between electricity and gas prices. If electricity is heavily taxed or burdened by network charges while gas remains relatively cheaper, the private incentive to switch is weakened even when the social and energy-security case is strong. In rented housing, the split-incentive problem can also arise: the landlord pays for the investment, while the tenant benefits from lower bills.

This mini-case shows the chapter's argument in practical terms. Heat pumps are efficient and dependence-reducing in physical terms, but their diffusion depends on relative prices, financing conditions, building characteristics and policy design. It also depends on whether the additional electricity consumed is increasingly domestic and low-carbon.

Electric vehicles present a similar logic. They can reduce oil dependence because they replace gasoline and diesel with electricity. This is particularly important for Italy because transport is the most oil-dependent sector. However, EVs require charging infrastructure, distribution-grid capacity and smart charging rules. If charging is concentrated during evening peaks, EVs can increase system stress. If charging is flexible, they can become an asset for the system, absorbing renewable electricity when it is abundant and reducing peak pressure.

## 6.4. Main Limits and Trade-Offs

The first limit is electricity demand. Electrification increases load and may increase peak demand. This requires investment in generation, transmission, distribution and flexibility. Heat pumps and electric vehicles are valuable technologies, but they do not provide the necessary grid capacity by themselves.

The second limit is timing. Boilers, cars and industrial equipment are durable assets. Their replacement takes time because households and firms do not change them every year. This means that electrification has high long-term potential, but its effects depend on replacement cycles, incentives and infrastructure availability.

The third limit is the electricity mix. Electrification reduces dependence only if the electricity system becomes progressively cleaner and more domestic. On a fossil-heavy grid, electrification may reduce direct fossil-fuel use in buildings and transport, but it can also increase fossil-fuel use in power generation. This is why electrification and renewable deployment must proceed together.

The fourth limit is affordability. Heat pumps and electric vehicles often have higher upfront costs than conventional alternatives. Even if operating costs are lower over time, households and firms may be constrained by liquidity, uncertainty or lack of information. Policy must therefore address the same barriers discussed in Chapter 5: capital constraints, information problems, split incentives and behavioural inertia.

The fifth limit is regulation. Relative prices matter. If electricity taxation and tariff structures make electricity artificially expensive compared with gas, the private incentive to electrify becomes weaker. A credible

electrification strategy therefore requires tariff reform, targeted support for vulnerable households, and price signals that encourage flexible consumption.

## 6.5. Implications for Reducing Energy Dependence

Electrification is one of the most important instruments for reducing Italy's dependence on imported gas and oil. It is particularly relevant for two sectors: heating and transport. Heat pumps can reduce gas demand in buildings, while electric vehicles can reduce oil demand in road transport. These are two of the main channels through which international energy markets affect Italian households and firms.

However, electrification is conditional. Its contribution depends on the evolution of the electricity system. If additional electricity demand is increasingly supplied by domestic renewables, electrification reduces import dependence and emissions at the same time. If the additional demand is supplied by gas-fired generation or imported electricity, the reduction in dependence is weaker. For this reason, electrification must be combined with renewable deployment, grid investment, storage, interconnections and demand-side flexibility.

Efficiency and electrification should therefore be treated as a sequence. Efficiency comes first because it reduces the amount of energy service that must be supplied. Electrification then shifts the remaining demand from imported fossil fuels to electricity. Renewables and flexibility make that electricity increasingly domestic, low-carbon and reliable. In this portfolio, electrification is not a stand-alone solution, but it is a central bridge between demand-side policy and supply-side transformation.

## 6.6. Comparison Table

The following matrix summarises the main demand-side options discussed in Chapters 5 and 6. The three options should be read as complements, not substitutes: efficiency reduces the load, while heat pumps and electric vehicles electrify the remaining demand.

| **Criterion** | **Energy efficiency** | **Heat pumps** | **Electric vehicles** |
|---|---|---|---|
| Import reduction | High and broad: reduces gas, oil and electricity demand at source. The energy not used requires no fuel. | High for gas: replaces gas burned for heating with electricity. | High for oil: replaces gasoline and diesel in road transport with electricity. |
| Price exposure | Reduces exposure in proportion to the demand avoided. | Reduces gas-price exposure but increases exposure to electricity prices. | Reduces oil-price exposure but increases exposure to electricity prices. |
| Environmental impact | Strong and immediate because avoided energy use avoids combustion. | Strong and increasing as the electricity mix decarbonises. | Strong and increasing as charging is supplied by renewables. |

| | | | |
|---|---|---|---|
| Cost and investment | Many measures can be cost-effective, but upfront costs and poor subsidy design are important barriers. | Higher upfront cost than gas boilers; payback depends on electricity-gas price ratios and building characteristics. | Higher purchase cost than conventional vehicles, though declining; requires charging infrastructure. |
| Implementation time | Medium to long term, because buildings and equipment are long-lived. | Medium term, limited by boiler replacement cycles, installer capacity and building suitability. | Medium term, limited by vehicle fleet turnover and charging rollout. |
| Scalability in Italy | Large potential, especially in buildings and industry, but limited by financing and split incentives. | Large potential in detached and low-rise buildings; more complex in dense condominiums and poorly insulated homes. | Large potential in passenger transport; constrained by charging access and distribution grids. |
| System integration | Positive: reduces peak load and network stress. | Adds winter electricity demand and requires smart controls and tariff reform. | Can add evening peaks if unmanaged; smart charging can turn EVs into a flexibility resource. |

The demand-side answer to the research question is therefore clear. Efficiency is the first and most robust lever because the molecule never demanded is the molecule never imported. Electrification is the structural substitution that can detach heating and transport from imported gas and oil. However, both depend on policy design. Their effectiveness is limited by capital constraints, split incentives, tariff structures and the carbon intensity of electricity. The portfolio, not the single technology, is the answer: efficiency plus electrification can reduce import-exposed demand significantly, but only if the rest of the system supplies the additional electricity through domestic, low-carbon and flexible sources.

# *7. Renewable Electricity: Solar and Wind as Domestic Supply*

## 7.1. Brief introduction

In the Italian case, reducing energy dependence is not only a matter of producing more energy domestically. It is also a matter of replacing imported primary fuels with domestic, low-carbon energy flows and making those flows usable within the electricity system. This is why solar and wind are central to the strategy. Once installed, they generate electricity without requiring continuous imports of gas, oil or coal. They can therefore reduce the exposure of the Italian electricity system to international fossil-fuel markets.

However, renewables are not a complete solution on their own. Solar and wind have low marginal costs, but their output is variable, weather-dependent and location-specific. Their contribution to energy security depends not only on how much capacity is installed, but also on when electricity is produced, where it is produced and whether the grid can absorb it. For this reason, renewable electricity must be analysed as part of a system strategy, together with flexibility, storage, transmission and market design.

## 7.2. Economic analysis

Solar and wind reduce dependence on imported fuels because they convert domestic natural resources into electricity. Their marginal cost is very low because they do not require fuel purchases. This is important for Italy because a significant part of electricity prices is still influenced by gas-fired generation. When renewables displace gas generation, they reduce both fuel imports and exposure to gas price volatility.

The short-run mechanism is straightforward. Renewable generation shifts the electricity supply curve to the right. When solar or wind production increases, more low-marginal-cost electricity enters the market and fossil generation is displaced. This can lower wholesale prices in the hours in which renewables are abundant. In this sense, renewable electricity can improve affordability as well as sustainability.

However, the welfare analysis is more complex. Solar and wind have high upfront capital costs and low operating costs. Their economic performance therefore depends on the cost of capital, permitting risk, expected revenues and long-term price signals. If regulation is uncertain or authorization procedures are slow, investors require higher returns or delay projects. This increases the cost of the transition.

Another important point is that renewables have system costs. These include grid expansion, balancing services, curtailment management, reserve capacity and congestion management. These costs do not mean that renewables are inefficient. They mean that the correct comparison is not between one solar plant and one gas plant in isolation. The correct comparison is between alternative system architectures: a fossil-based system exposed to fuel imports and a renewable-based system requiring more flexibility and network investment.

For this reason, the value of renewable electricity depends on timing and location. Solar generation is especially valuable when it coincides with daytime demand, but less valuable if many plants produce simultaneously and create oversupply. Wind can be complementary because its production profile is different from solar, but it is also geographically constrained. A balanced renewable portfolio therefore has higher system value than a strategy based on one technology alone.

## 7.3. Application to Italy

Italy has strong reasons to expand renewable electricity. The country has limited domestic fossil resources, high dependence on imported gas and oil, and good renewable potential, especially for solar PV. In 2024, Italy’s electricity demand was about 312 TWh, and renewable sources covered a record share of national electricity demand. At the same time, Italy remained a net electricity importer and natural gas still played a major role in power generation. This means that renewable expansion is not only a climate policy, but also an energy-security policy.

Solar PV is particularly relevant for Italy. The country has favourable solar irradiation, a large stock of rooftops and potential for both distributed and utility-scale projects. Solar can directly reduce gas-fired electricity generation during daylight hours. It is also compatible with electrification: as heat pumps and electric vehicles increase electricity demand, domestic solar generation can reduce the fossil-fuel content of that additional demand.

Wind has a complementary role. Although Italy's wind potential is more geographically concentrated than solar, especially in southern and insular regions, wind can produce at different times and seasons. This diversification matters because a renewable system based only on solar would create stronger daily concentration of production. Wind therefore improves the temporal and geographic diversification of domestic renewable supply.

However, Italy also shows why renewable capacity alone is not enough. Renewable potential is not always located close to demand centres. New projects in southern regions or islands reduce dependence only if electricity can be transported to the areas where it is needed. If transmission or distribution networks are congested, renewable output may be curtailed. In that case, Italy would have installed domestic clean capacity but would not fully convert it into usable energy.

Permitting is another major constraint. Slow authorization procedures increase uncertainty, delay investment and raise costs. From an economic perspective, permitting delays act like an additional cost of capital: they make renewable projects riskier and reduce the speed at which domestic supply can replace imported fuels. For Italy, administrative simplification is therefore not a secondary issue, but a core energy-security instrument.

## 7.4. Main limits and trade-offs

The first limit is intermittency. Solar and wind reduce imported fuel use over the year, but they do not guarantee production in every hour. The system still needs flexibility when renewable output is low and must manage oversupply when renewable output is high.

The second limit is grid integration. Renewables reduce dependence only if their electricity reaches consumers. Transmission expansion, distribution upgrades and digital grid management are therefore necessary complements. Without them, additional capacity can lead to congestion and curtailment.

The third limit is affordability. Although renewables have low marginal costs, they require large upfront investments. These investments must be financed through private capital, public support, electricity tariffs or a combination of instruments. Poorly designed support schemes can increase costs for consumers, while well-designed auctions and stable long-term rules can lower financing costs.

The fourth limit is market design. In a system with high shares of low-marginal-cost renewables, wholesale prices can become very low in some hours and very high in others. This creates investment uncertainty for both renewable plants and flexibility resources. Market rules must therefore reward not only energy production, but also capacity, flexibility, location and reliability.

The fifth limit is social acceptance. Utility-scale solar, wind farms and transmission lines can face local opposition. This does not eliminate their economic value, but it means that policy must manage distributional conflicts, land-use concerns and benefit-sharing mechanisms.

## 7.5. Implications for reducing energy dependence

Renewable electricity is one of the strongest supply-side instruments for reducing Italy's energy dependence. Solar and wind directly reduce the need for imported fossil fuels in power generation. They also make electrification more effective, because heat pumps and electric vehicles reduce import dependence only if the additional electricity is increasingly produced from domestic low-carbon sources.

Using the criteria defined in Chapter 4, renewables perform strongly on import reduction, environmental impact and long-term scalability. Their main weaknesses are not fuel cost or technological maturity, but

integration, permitting, investment timing and system coordination. In other words, the bottleneck is not only "how much renewable capacity Italy can build", but "how much renewable electricity the system can absorb, deliver and value correctly".

The implication for the research question is clear. Renewables are necessary but not sufficient. They provide the domestic low-carbon supply needed to reduce external dependence, but their full contribution is realized only if Italy also invests in grids, storage, demand response, interconnections and appropriate market rules. This prepares the next chapter, which analyses the flexibility infrastructure required to make a high-renewables system reliable and economically efficient.

## 7.6. Comparison table

| Option | Impact on energy dependence | Time horizon | Relative cost | Main limits |
|---|---|---|---|---|
| Solar PV | High, because it replaces imported fossil generation with domestic electricity, especially during daytime hours | Short to medium term | Low marginal cost, high upfront CAPEX | Intermittency, land use, permitting, grid congestion |
| Wind power | High, especially as a complement to solar and in southern/insular areas | Medium term | Low marginal cost, high upfront CAPEX | Location constraints, permitting, social acceptance, variability |
| Renewable auctions | Indirect but important, because they reduce financing costs and support deployment | Short to medium term | Potentially cost-efficient if well designed | Risk of underbidding, administrative complexity, regulatory uncertainty |
| Distributed generation | Moderate to high, especially for rooftop PV and self-consumption | Short to medium term | Falling costs, but still investment-intensive for households and firms | Distribution-grid constraints, tariff design, unequal access |
| Utility-scale renewables | High, because they can deliver large volumes of domestic electricity | Medium term | Economies of scale | Transmission needs, land-use conflicts, authorization delays |

| | | | | |
|---|---|---|---|---|
| Renewable electricity plus electrification | Very high, because it reduces gas and oil use in heating and transport | Medium to long term | Requires coordinated investment | Depends on grid capacity, flexibility and electricity demand management |

# *8. Grid, Storage, Interconnections and System Flexibility*

## 8.1. Brief introduction

Chapter 7 showed that solar and wind can reduce Italy's dependence on imported fossil fuels by increasing domestic low-carbon electricity production. However, renewable generation alone is not sufficient. In a high-renewables system, the binding constraint is often not only generation capacity, but the ability to move electricity across space, balance it over time and value flexibility correctly.

For this reason, grids, storage, interconnections and demand response are not secondary technologies. They are the infrastructure that makes domestic renewable electricity usable, reliable and economically valuable. Without flexibility, Italy may install more renewable capacity but still depend on gas-fired generation or imports in hours when renewable output is low. With flexibility, the same renewable capacity can displace more imported fossil fuels and contribute more effectively to energy security.

## 8.2. Economic analysis

The economic logic is simple. Intermittent renewables lower the marginal cost of electricity generation, but they increase the value of flexibility. When supply varies with weather and time of day, the system needs resources that can shift electricity across hours, between regions and between peak and off-peak demand.

There are three main flexibility channels.

The first channel is spatial flexibility. Electricity can be moved across space through transmission lines and interconnectors. This is important because renewable generation and electricity demand are not always located in the same places. In Italy, renewable potential is often stronger in southern regions and islands, while demand is concentrated in industrial and urban areas. Grid reinforcement therefore increases the value of renewable generation by allowing domestic clean electricity to reach consumers.

Interconnections are part of the same logic at the European level. They allow countries to share balancing resources, reduce reserve requirements and exploit geographic diversity. When solar or wind generation is low in one area, it may be higher elsewhere. Interconnectors can also reduce price volatility by allowing electricity to flow from lower-price to higher-price areas. In this sense, cross-border electricity trade is not necessarily a weakness. If well managed, it can be a flexibility asset. The problem arises when imports become a structural substitute for domestic capacity rather than a balancing mechanism.

The second channel is temporal flexibility. Storage shifts electricity over time. Batteries can absorb electricity when renewable generation is abundant and release it during peak demand. They are particularly useful for short-duration services such as frequency response, peak shaving, congestion management and the management of electric-vehicle charging. However, batteries should not be presented as a universal solution. For long-duration or seasonal balancing, other options such as pumped hydro, hydro reservoirs, demand response, flexible generation or sector coupling may be more appropriate.

The third channel is demand-side flexibility. Demand response reduces the need for physical backup capacity by making consumption more responsive to system conditions. Smart meters, digital platforms and dynamic tariffs make it possible to shift some electricity demand away from peak hours or toward periods of high renewable generation. Electric vehicles, heat pumps and industrial loads can become flexibility resources if they are controlled through appropriate price signals and automation.

The economic value of flexibility is therefore not limited to avoiding blackouts. Flexibility reduces the total cost of integrating renewables. It lowers curtailment, reduces congestion, limits the need for expensive peak generation and increases the capacity of renewable electricity to replace fossil fuels. In this sense, flexibility increases the system value of renewables.

Market design is essential for this process. If electricity prices do not reflect scarcity, congestion or time variation, consumers and investors receive weak signals. Storage operators may not be rewarded for the services they provide. Demand response may remain unused. Renewable projects may be located where the resource is strong but the grid is weak. A high-renewables system therefore requires more granular price signals, efficient network tariffs and markets that reward capacity, balancing and ancillary services.

## 8.3. Application to Italy

Italy is a particularly relevant case because the energy transition increases the centrality of the electricity system. Electrification of heating and transport will increase electricity demand and change its daily and seasonal profile. At the same time, the expansion of solar and wind will make electricity supply more variable. The combination of these two trends makes flexibility a strategic requirement.

As shown in Chapter 3, Italy remains a net electricity importer and natural gas still plays an important role in power generation. This means that flexibility is directly linked to energy dependence. If Italy cannot use domestic renewable electricity efficiently, it will continue to rely on gas-fired generation and cross-border imports in critical hours. If it improves networks, storage, interconnections and demand response, each additional unit of renewable generation will reduce imported fossil fuels more effectively.

Transmission and distribution grids are the first priority. Utility-scale solar and wind projects are often located far from the main demand centres. Rooftop solar, heat pumps and electric vehicles create new pressures on distribution networks. Therefore, the grid must evolve from a passive delivery infrastructure into an active platform capable of managing decentralised generation, flexible loads and bidirectional flows.

Storage is the second priority. In the short run, batteries can help manage daily solar variability, local congestion and peak demand. They are especially relevant in combination with solar PV, because they can shift midday generation toward evening demand. However, storage deployment must be guided by system value, not only by installed capacity targets. Batteries should be placed where they reduce congestion, improve reliability or support flexible demand.

Interconnections are the third priority. Italy is part of the European electricity system, and cross-border trade can improve security by diversifying supply conditions. Interconnections can reduce the cost of balancing renewables and provide access to different generation profiles across Europe. However, they should complement domestic renewable deployment and flexibility, not replace them. In the context of energy dependence, the goal is not isolation from European markets, but lower structural vulnerability to imported fossil fuels.

Demand response is the fourth priority. Electrification creates new flexible loads. Electric vehicles can charge when renewable generation is abundant or prices are low. Heat pumps can be combined with thermal storage and smart controls. Industrial consumers can shift some processes if they are rewarded through appropriate contracts. These options reduce the need for fossil backup and make electrification less costly for the system.

## 8.4. Main limits and trade-offs

The first limit is investment cost. Grid reinforcement, digitalisation, storage and interconnections require large capital expenditure. These costs must be recovered through tariffs, public support or market revenues. If cost recovery is poorly designed, the transition may become less affordable or socially contested.

The second limit is timing. Grid projects and interconnections often have long lead times because they require planning, permitting, land use decisions and coordination between institutions. Renewable plants can sometimes be built faster than the networks needed to connect them. This creates a risk of bottlenecks and curtailment.

The third limit is regulation. Flexibility does not emerge automatically. It depends on market rules that allow storage, demand response and flexible consumers to participate in electricity markets. If tariffs are flat,

consumers have little incentive to shift demand. If ancillary services are poorly priced, storage and flexible resources may not receive adequate revenues.

The fourth limit is distributional. Network tariffs and dynamic pricing can create winners and losers. Consumers with smart technologies may benefit from flexibility, while vulnerable households may have less ability to adjust consumption. Policy design must therefore combine efficiency with fairness.

The fifth limit is coordination. Flexibility requires coordination between transmission system operators, distribution system operators, regulators, electricity retailers, consumers and investors. Without coordination, investments may be delayed or located in the wrong places.

## 8.5. Implications for reducing energy dependence

Grid flexibility does not reduce fossil fuel imports in the same direct way as energy efficiency or renewable generation. However, it is essential because it determines how much renewable electricity can actually replace imported gas and oil. A strong renewable build-out without grids, storage and demand response risks producing stranded clean electricity in some hours and continued fossil dependence in others.

By contrast, a coordinated flexibility strategy allows Italy to replace imported fuels more consistently across time and space. A better grid lowers curtailment. Interconnections reduce reserve requirements and price volatility. Storage absorbs short-term variability. Demand response smooths peaks created by electrification. Together, these measures make each additional megawatt of renewable capacity more effective in reducing energy dependence.

For this reason, the reduction of energy dependence should be measured at system level. The objective is not simply to install more renewable capacity, but to transform renewable output into firm, deliverable and valuable energy services. Flexibility is what turns renewable capacity into real energy security.

## 8.6. Comparison table

| Option | Impact on energy dependence | Time horizon | Relative cost | Main limits |
|---|---|---|---|---|
| Grid reinforcement | Indirect but essential, because it allows domestic renewable generation to reach demand centres | Medium to long term | High investment cost | Long lead times, permitting, local opposition, tariff impact |
| Battery storage | Moderate to high for short-term balancing, peak shaving and congestion management | Short to medium term | Falling costs, but still costly for long-duration use | Limited duration, revenue uncertainty, location-specific value |
| Pumped hydro and hydro reservoirs | High system value for medium- and long-duration flexibility where available | Medium to long term | High upfront cost | Geographic constraints, environmental permitting, long development times |

| | | | | |
|---|---|---|---|---|
| Interconnections | High system value, because they reduce variability and allow reserve sharing across borders | Medium to long term | High but system-efficient | Cross-border coordination, investment risk, dependence on external market conditions |
| Demand response | High enabling value, because it reduces peak demand and makes electrification easier to integrate | Short to medium term | Relatively low compared with physical infrastructure | Requires smart meters, dynamic tariffs, automation and consumer participation |
| Smart charging for EVs | High potential, because it turns new electric load into a flexibility resource | Short to medium term | Moderate, linked to digital and charging infrastructure | Behavioural constraints, tariff design, charger availability |
| Market design reform | Indirect but central, because it rewards flexibility, location and reliability | Medium term | Low physical cost but high institutional complexity | Political resistance, regulatory complexity, distributional effects |

The connection to the research question is clear. The best way to reduce Italian energy dependence is not to choose between generation and flexibility, but to treat them as complements. Renewables create the domestic supply base; grids, storage, interconnections and demand response convert that supply into firm system value. In the logic of this paper, flexibility is what makes the renewable transition credible, scalable and economically efficient in Italy.

# *9. Biomethane, Hydrogen and Hard-to-Electrify Sectors*

## 9.1. Brief introduction

The previous chapters have shown that the main pillars for reducing Italy's energy dependence are energy efficiency, electrification, renewable electricity and system flexibility. However, not all final uses can be electrified easily or economically. Some industrial processes, high-temperature heat applications, chemical feedstocks and segments of heavy transport may require energy carriers other than direct electricity.

This is where biomethane and hydrogen become relevant. Their role should not be exaggerated: they are not universal substitutes for natural gas, oil or electricity. Their economic value is narrower but still important. They can contribute to reducing import dependence in specific sectors where direct electrification is technically difficult, costly or inefficient.

The key argument of this chapter is that biomethane and hydrogen should be treated as targeted solutions. They are useful when they replace imported fossil fuels in hard-to-electrify uses, but they should not absorb scarce public support in sectors where efficiency, direct electrification or renewable electricity are already cheaper and more efficient.

## 9.2. Economic analysis

Biomethane is attractive because it can partially substitute imported natural gas while using parts of the existing gas infrastructure. This makes it relevant for a country like Italy, where gas still plays an important role in heating, industry and power generation. In principle, biomethane can reduce dependence without requiring a complete redesign of end-use technologies.

However, the main economic constraint is scale. Biomethane production depends on sustainable feedstocks such as agricultural residues, organic waste, manure and other biomass sources. These inputs are limited and may face competing uses. This means that biomethane should be understood as a valuable complement, not as a full replacement for imported natural gas.

The policy implication is selective use. Biomethane creates the highest system value when it replaces fossil gas in uses that are difficult to electrify, or where gas infrastructure already exists and the marginal cost of switching is low. It is less efficient to use scarce biomethane in applications that could be served by heat pumps or direct renewable electricity at lower cost.

Hydrogen has a different economic profile. It is an energy carrier, not a primary source. This means it must be produced using another form of energy. Grey hydrogen is produced from fossil fuels and is emissions-intensive. Blue hydrogen can reduce emissions if carbon capture is effective, but it still depends on natural gas and carbon-storage infrastructure. Green hydrogen has the strongest environmental profile, but it requires large amounts of renewable electricity and remains expensive compared with direct electrification in many uses.

This creates an opportunity-cost problem. Every unit of renewable electricity used to produce hydrogen is a unit that cannot be used directly for heat pumps, electric vehicles or industrial electrification. Since direct use of electricity is usually more efficient, green hydrogen should be reserved for uses where direct electrification is technically difficult or where hydrogen is needed as a chemical input.

The economics of hydrogen are also shaped by transport and storage costs. Hydrogen has high energy content by weight but low energy density by volume, which makes compression, storage and transportation costly. This limits its role as a general-purpose replacement for natural gas. It is more credible as a targeted energy carrier for industrial clusters, ports, refineries, chemicals, steel-related processes and some heavy transport segments.

## 9.3. Application to Italy

For Italy, biomethane is the more immediately plausible option. It can connect with the existing gas system and reduce imported natural gas in selected uses. This matters because Italy's energy system remains gas-intensive in buildings, industry and electricity generation. In the short run, biomethane can reduce gas-import dependence more directly than hydrogen because it can be injected into the gas grid and used with limited adaptation by end users.

However, the scale limitation remains decisive. The sustainable biomass resource available in Italy is finite. Therefore, biomethane cannot be presented as the solution to national gas dependence. Its most credible role is to provide a domestic renewable gas for selected uses, especially where electrification is difficult, where local feedstocks are available, or where integration with agriculture and waste management creates additional environmental benefits.

Hydrogen in Italy should be understood as a solution for hard-to-electrify niches rather than as a substitute for gas in residential heating or passenger mobility. The most plausible applications are industrial processes, high-temperature heat, chemical uses, refineries, some heavy transport segments and possibly ports and logistics hubs. This fits the logic of the project: hydrogen should be used where it avoids the highest-cost alternatives, not where efficiency or direct electrification is already cheaper and more mature.

The Italian policy question is therefore not whether biomethane or hydrogen "work" in general. The real question is where they can reduce import dependence at acceptable cost and with realistic infrastructure requirements. If hydrogen is used in sectors that could be electrified directly, it may increase total electricity demand and system costs without delivering proportional security benefits. If it is targeted to industrial clusters and hard-to-electrify processes, it can become a useful complement to the broader transition.

## 9.4. Main limits and trade-offs

The first limit is scale. Biomethane cannot replace all imported natural gas because sustainable feedstocks are limited. Hydrogen cannot replace all fossil fuels because it is energy-intensive to produce, costly to transport and inefficient in many applications compared with direct electrification. Both fuels are useful only if policy avoids overpromising their potential.

The second limit is cost. Biomethane is generally more expensive than fossil natural gas, and green hydrogen is still more expensive than direct use of electricity in most applications. This creates a clear prioritisation problem: public support should be directed to the sectors where these fuels generate the highest import-reduction and decarbonisation value.

The third limit is infrastructure. Biomethane requires upgrading plants, grid injection points, sustainability certification and quality standards. Hydrogen requires electrolysers, dedicated transport and storage systems, safety regulation and demand-side adaptation. These infrastructures take time and capital, and they may create lock-in risks if built before demand is sufficiently clear.

The fourth limit is sustainability. Biomethane is only environmentally valuable if feedstocks are sustainable and if methane leakage is controlled. Hydrogen is only low-carbon if its production process is low-carbon. Grey hydrogen does not solve the climate problem, while blue hydrogen depends on the effectiveness of carbon capture and storage. Green hydrogen depends on the availability of additional renewable electricity.

The fifth limit is governance. These fuels require coordinated planning between energy policy, industrial policy, agriculture, waste management and infrastructure regulation. Without coordination, there is a risk of fragmented projects, inefficient subsidies and weak contribution to energy security.

## 9.5. Implications for reducing energy dependence

Biomethane and hydrogen can reduce Italy's energy dependence, but only in a selective way. Their role is not to replace the main pillars of the transition. Efficiency, renewables, electrification and flexibility remain the

core instruments because they address large parts of final demand and domestic supply. Biomethane and hydrogen are complementary instruments for the segments where those main options are limited.

Using the criteria defined in Chapter 4, biomethane performs relatively well on short- to medium-term feasibility because it can use existing gas infrastructure. Its import-reduction impact is positive but limited by feedstock availability. Hydrogen has higher long-term potential in hard-to-electrify sectors, but it performs less strongly on affordability and implementation time because of high costs, infrastructure needs and conversion losses.

The policy implication is prioritisation. Scarce renewable gases should be allocated to the uses with the highest system value: industrial processes, chemical feedstocks, heavy transport niches and other applications where direct electrification is technically or economically constrained. They should not be used as a general justification for delaying efficiency improvements, electrification or renewable deployment.

The connection to the research question is therefore clear. Biomethane and hydrogen contribute to reducing Italy's energy dependence only if they are treated as targeted complements. They are valuable where they replace imported fossil fuels in difficult sectors, but they are not the main answer to Italy's vulnerability. The main answer remains a coordinated portfolio: reduce demand through efficiency, electrify where efficient, produce more domestic renewable electricity, strengthen flexibility and use renewable gases only where they create the highest additional value.

## 9.6. Comparison table

| Option | Impact on energy dependence | Time horizon | Relative cost | Main limits |
|---|---|---|---|---|
| Biomethane | Medium: can replace imported natural gas in selected uses and can use existing gas infrastructure | Short to medium term | Higher than fossil gas, but potentially lower adaptation cost than hydrogen | Limited sustainable feedstocks, methane leakage risks, certification needs |
| Green hydrogen | Medium to high in specific hard-to-electrify sectors, low as a general-purpose fuel | Medium to long term | High, due to electrolysis, renewable electricity needs, transport and storage | High cost, conversion losses, infrastructure needs, competition for renewable electricity |
| Blue hydrogen | Potentially medium where carbon capture is effective | Medium term | Lower than green hydrogen in some cases, but dependent on gas and CCS | Fossil-gas dependence, carbon-capture uncertainty, storage infrastructure |

| | | | | |
|---|---|---|---|---|
| Industrial hydrogen use | High system value when used as feedstock or for high-temperature processes | Medium to long term | High but potentially justified in hard-to-abate sectors | Requires industrial clusters, stable demand and dedicated infrastructure |
| Hydrogen for passenger mobility | Low in most cases because battery-electric vehicles are more efficient | Medium term | High | Infrastructure duplication, lower efficiency than direct electrification |
| Biomethane in buildings | Low to medium: useful in limited cases, but heat pumps and efficiency often have higher system value | Short to medium term | Medium to high | Risk of using scarce biomethane where electrification is feasible |
| Renewable gases for power backup | Medium as a reserve option for system adequacy | Long term | High | Should be limited to backup or flexibility, not baseload use |

# *10. Final Comparative Assessment: Which Options Reduce Italy's Energy Dependence the Most?*

The previous chapters analysed the main technological and policy options available to Italy: energy efficiency, electrification, renewable electricity, grid flexibility, storage, interconnections, biomethane and hydrogen. This chapter compares them using the criteria defined in Chapter 4. The objective is not to identify a single "best" technology, but to understand which options reduce energy dependence directly, which ones enable other measures, and which ones should be used only in specific sectors.

The comparison confirms the main argument of the paper: Italy's energy dependence cannot be reduced through one technology alone. The Italian energy system is vulnerable through several channels: oil in transport, gas in buildings and industry, gas in electricity generation, and electricity imports during specific periods. For this reason, each option addresses a different part of the problem.

Energy efficiency reduces the size of final energy demand. Renewables increase domestic low-carbon electricity supply. Electrification replaces imported gas and oil in heating and transport. Grid flexibility makes renewable electricity usable and reliable. Biomethane and hydrogen provide complementary solutions for hard-to-electrify sectors.

## 10.1. Comparative matrix

The matrix below evaluates each option according to the eight criteria introduced in Chapter 4. The ranking uses three qualitative levels: **High**, **Medium** and **Low**. A high score does not always mean that the option directly reduces imports; in some cases, it means that the option is essential for enabling other measures.

| Option | Import reduction | Energy security | Affordability | Implementation time | Scalability | Environmental impact | System integration | Regulatory feasibility |
|---|---|---|---|---|---|---|---|---|
| **Energy efficiency** | High | High | High | Medium | High | High | High | Medium |
| **Renewable electricity** | High | High | Medium-High | Medium | High | High | Medium | Medium |
| **Heat pumps** | High for gas | High | Medium | Medium | Medium-High | High | Medium | Medium |
| **Electric vehicles** | High for oil | High | Medium | Medium-Long | Medium | High | Medium | Medium |
| **Grid reinforcement** | Indirect but essential | High | Medium | Long | High | Medium-High | High | Medium-Low |
| **Battery storage** | Indirect | Medium-High | Medium | Medium | Medium | Medium-High | High | Medium |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| **Interconnections** | Indirect | Medium-High | Medium-High | Long | Medium | Medium | High | Medium-Low |
| **Demand response** | Indirect but important | Medium-High | High | Short-Medium | Medium-High | Medium-High | High | Medium |
| **Biomethane** | Medium | Medium | Medium-Low | Medium | Low-Medium | Medium-High | Medium | Medium |
| **Green hydrogen** | Medium in specific sectors | Medium-High long term | Low-Medium | Long | Low-Medium | High if renewable | Medium-Low | Low-Medium |

The matrix shows that the strongest options are not necessarily those with the most visible technological profile. Energy efficiency performs very strongly because it reduces dependence at the source. It lowers the amount of imported energy required to provide heating, mobility, industrial output and other energy services. It also reduces pressure on the electricity system, making electrification and renewable integration easier.

Renewable electricity is the second major structural pillar. Solar and wind directly reduce the need for imported fossil fuels in electricity generation and make electrification more effective. However, their final contribution depends on grid capacity, storage, market design and permitting procedures. Renewables are therefore necessary, but their impact is constrained by system integration.

Electrification is highly strategic, but conditional. Heat pumps can reduce gas dependence in buildings, while electric vehicles can reduce oil dependence in transport. However, both increase electricity demand. Their energy-security value depends on whether the additional electricity is supplied by domestic renewables and whether the grid can manage the new load.

Grid reinforcement, storage, interconnections and demand response do not reduce imports in the same direct way. Their role is enabling. They increase the amount of renewable electricity that can be absorbed, transported and consumed. Without them, renewable capacity may face congestion, curtailment and lower economic value. In this sense, flexibility transforms renewable capacity into real energy security.

Biomethane and hydrogen have a narrower role. Biomethane can reduce gas imports in selected uses, but its potential is limited by sustainable feedstock availability. Hydrogen has potential in hard-to-electrify sectors, but high costs, conversion losses and infrastructure requirements make it unsuitable as a general-purpose solution. Both should be targeted to sectors where direct electrification is technically or economically difficult.

## 10.2. Direct, enabling and targeted measures

The comparison suggests that the options can be divided into three categories: direct measures, enabling measures and targeted measures.

**Direct measures** are those that reduce dependence by lowering energy demand or replacing imported fuels. Energy efficiency, renewable electricity and electrification belong to this category. Efficiency reduces the amount of energy needed. Renewables replace fossil generation with domestic low-carbon electricity. Electrification replaces gas and oil in final uses.

**Enabling measures** do not necessarily reduce dependence immediately, but they determine whether direct measures can scale. Grid reinforcement, storage, interconnections and demand response are enabling measures.

Their value lies in increasing the reliability and economic efficiency of a high-renewables and electrified system.

**Targeted measures** are useful only in specific sectors. Biomethane and hydrogen belong to this category. They should not be used as substitutes for the broader transition. Their role is to address applications where efficiency, direct electrification and renewable electricity are not sufficient.

This classification is important because it avoids a common policy mistake: treating all technologies as if they were substitutes. In reality, most options are complements. Efficiency reduces the size of the problem; electrification changes the structure of final demand; renewables provide domestic electricity; flexibility makes that electricity reliable; and renewable gases cover residual difficult sectors.

## 10.3. Ranking of the options

Based on the criteria used in the matrix, the options can be ranked according to their contribution to reducing Italy's energy dependence.

**First priority: energy efficiency.**
Energy efficiency should be the first policy priority because it reduces demand before the system needs to supply it. It is the most robust option across the energy trilemma: it improves security, reduces emissions and can lower costs over time. Its main barriers are not technological, but financial, behavioural and regulatory. For Italy, the largest potential is in buildings and industry, where efficiency directly reduces gas demand and price exposure.

**Second priority: renewable electricity.**
Solar and wind are the main supply-side instruments for reducing dependence. They increase domestic low-carbon production and reduce the role of imported fossil fuels in electricity generation. Their long-term potential is very high, but their effectiveness depends on permitting reform, grid expansion and market design. Renewable electricity is also necessary to make electrification a genuine energy-security strategy.

**Third priority: electrification of final uses.**
Electrification is essential for sectors where fossil fuels dominate final consumption. Heat pumps reduce gas dependence in buildings; electric vehicles reduce oil dependence in transport. However, electrification must be sequenced correctly. It is strongest when combined with efficiency first and renewables after. Electrifying inefficient buildings or unmanaged vehicle charging can create new pressure on the electricity system.

**Fourth priority: grid flexibility, storage and demand response.**
These measures are not optional. They are the condition for scaling renewables and electrification. Their impact on import dependence is indirect, but without them Italy cannot fully exploit domestic renewable resources. Grid investment, storage and flexible demand reduce the need for gas backup, lower curtailment and improve reliability.

**Fifth priority: biomethane and hydrogen.**
Biomethane and hydrogen should be used selectively. They are valuable in hard-to-electrify sectors, but they cannot replace the central role of efficiency, renewables and electrification. Their policy support should be targeted to applications with high system value, such as industrial processes, chemical feedstocks, some heavy transport uses and backup applications.

## 10.4. Short-term versus long-term contribution

The time dimension changes the ranking.

In the **short term**, the most useful options are those that reduce demand or improve system operation without waiting for a complete transformation of infrastructure. Energy efficiency, demand response, smart energy management and accelerated renewable permitting can generate relatively fast benefits. These measures reduce exposure to imported fuels and price shocks without requiring full replacement of the existing capital stock.

In the **medium term**, renewable deployment, heat pumps, electric vehicles, batteries and distribution-grid upgrades become increasingly important. This is the period in which Italy can reduce gas and oil dependence structurally, provided that the electricity system expands in a coordinated way.

In the **long term**, the key challenge is system transformation. A highly electrified economy requires a much stronger grid, more storage, more interconnections, flexible demand and targeted low-carbon fuels for residual sectors. Hydrogen may become relevant in this long-term phase, but mainly in industrial and hard-to-electrify applications.

Therefore, the policy mix must be dynamic. Italy should not wait for long-term technologies before acting on efficiency and renewables. At the same time, it should not expand electrification without preparing the electricity system. Sequencing matters: first reduce demand, then electrify efficient uses, then supply them with domestic renewables, while building flexibility in parallel.

## 10.5. Main trade-offs

The comparison reveals five main trade-offs.

The first trade-off is between **speed and depth**. Energy efficiency and demand response can produce benefits relatively quickly, but deep structural change requires renewable deployment, electrification and grid investment. Short-term measures are necessary, but they are not sufficient.

The second trade-off is between **domestic supply and system integration**. Renewables increase domestic production, but their value depends on the system's ability to integrate them. Without grid capacity and flexibility, installed capacity does not automatically translate into reduced dependence.

The third trade-off is between **electrification and electricity security**. Electrification reduces gas and oil consumption, but it increases electricity demand. This creates a new dependency on the reliability, affordability and carbon intensity of the power system.

The fourth trade-off is between **cost and affordability**. Many transition technologies reduce operating costs over time, but they require upfront investment. If costs are not distributed fairly, policies may generate social resistance. This is especially relevant for building renovation, heat pumps, electric vehicles and network tariffs.

The fifth trade-off is between **broad solutions and targeted solutions**. Biomethane and hydrogen are useful, but only in selected sectors. Overusing them in applications where direct electrification is cheaper would increase system costs and reduce overall efficiency.

## 10.6. Overall result

The final result of the comparison is clear: the most effective strategy is not based on choosing one dominant technology, but on combining complementary measures.

The core strategy should be:

1. **reduce final energy demand through efficiency;**
2. **expand domestic renewable electricity;**
3. **electrify heating and transport where efficient;**
4. **strengthen grids, storage, interconnections and demand response;**
5. **reserve biomethane and hydrogen for hard-to-electrify sectors.**

This portfolio directly addresses Italy's main vulnerabilities. Efficiency reduces gas, oil and electricity demand at the source. Renewables reduce dependence on imported fuels in electricity generation. Electrification cuts direct use of gas in buildings and oil in transport. Flexibility ensures that renewable electricity can replace

fossil fuels reliably. Biomethane and hydrogen cover the residual parts of the system where direct electrification is less feasible.

The key message is that reducing energy dependence does not mean achieving full energy self-sufficiency. For Italy, this would be unrealistic and economically inefficient. The more credible objective is to reduce vulnerability: lower exposure to imported fossil fuels, lower sensitivity to international price shocks, lower emissions and a more resilient electricity system.

In this sense, the strongest policy conclusion is that efficiency and renewables are the two central pillars, electrification is the main substitution channel, and flexibility is the enabling infrastructure. Biomethane and hydrogen are useful complements, but only if they are targeted carefully. The Italian energy transition must therefore be designed as a coordinated system strategy rather than as a collection of isolated technology incentives.

# *11. Policy Implications: What Italy Should Do*

The comparative assessment developed in Chapter 10 shows that reducing Italy's energy dependence requires a coordinated policy package rather than isolated technological incentives. The policy objective should not be full energy self-sufficiency, which would be unrealistic and economically inefficient for Italy. The objective should be lower systemic vulnerability: less exposure to imported fossil fuels, lower sensitivity to international price shocks, lower emissions and a more resilient electricity system.

This chapter translates the previous analysis into policy implications. The central message is that Italy should prioritise measures that reduce demand, expand domestic renewable supply, electrify final uses where efficient, strengthen system flexibility and reserve renewable gases for sectors where direct electrification is difficult. The policy sequence matters: efficiency should reduce the size of the problem; renewables should provide domestic low-carbon electricity; electrification should shift final demand away from imported fossil fuels; flexibility should make the system reliable; and biomethane and hydrogen should be targeted to residual hard-to-electrify uses.

## 11.1. Put energy efficiency first

The first policy implication is that energy efficiency should be treated as the foundation of Italy's energy-security strategy. Efficiency reduces the need for imported energy before the system has to replace it with new supply. It also reduces pressure on electricity networks, makes electrification easier and lowers household and industrial exposure to volatile prices.

In buildings, policy should focus on deep but cost-controlled renovation. Italy should prioritise the worst-performing buildings, vulnerable households and public buildings, where efficiency gains can be large and socially valuable. The policy lesson from recent renovation incentives is not that subsidies are unnecessary, but that they must be targeted and fiscally sustainable. Support should be linked to measured energy performance, additionality and long-term savings, not only to expenditure.

A stronger efficiency policy should combine several instruments. Minimum energy performance standards can address split incentives in rented housing. Energy performance certificates and information tools can reduce information asymmetries. Concessional loans and guarantees can reduce liquidity constraints. Targeted grants can support low-income households. For firms, especially small and medium-sized enterprises, energy audits and investment support can accelerate process efficiency and reduce gas and electricity exposure.

The key point is that efficiency policy must be designed according to the "failure-matching" principle. Information failures require information instruments; liquidity constraints require finance; split incentives require standards or contractual solutions; environmental externalities require energy and carbon price signals. A generic subsidy alone is not enough.

## 11.2. Accelerate renewable deployment, but link it to system value

The second policy implication is that Italy should accelerate renewable electricity deployment, especially solar and wind. Renewables are the main domestic supply-side instrument for reducing dependence on imported fuels in electricity generation. They also make electrification meaningful: heat pumps and electric vehicles reduce fossil dependence only if the additional electricity is increasingly domestic and low-carbon.

The first priority is permitting reform. Slow authorisation procedures increase risk, delay projects and raise the cost of capital. From an energy-security perspective, administrative delay is equivalent to keeping Italy dependent on imported fuels for longer. Permitting should be faster, more predictable and better coordinated between national, regional and local authorities.

The second priority is grid-aware renewable planning. New renewable capacity should be encouraged where it has high system value, not only where the resource is strongest. Solar and wind projects should be

coordinated with transmission capacity, distribution-network needs, storage and demand centres. Otherwise, Italy risks installing capacity that cannot be fully absorbed, leading to congestion and curtailment.

The third priority is stable investment design. Auctions, contracts for difference or long-term procurement mechanisms can reduce financing costs, but they must avoid overprotecting investors from market signals. Support mechanisms should reward not only the quantity of renewable electricity generated, but also its contribution to system needs, including timing, location and reliability.

Distributed generation should also be encouraged, especially rooftop solar, energy communities and self-consumption. However, policy must avoid unequal access. Households without suitable roofs or investment capacity should not bear disproportionate network costs while others benefit from self-generation. Tariff design must therefore balance efficiency, fairness and cost recovery.

## 11.3. Electrify heating and transport with correct price signals

The third policy implication is that Italy should support electrification of final uses, especially heat pumps and electric vehicles, but only as part of a coordinated electricity-system strategy. Electrification is essential because it targets the two main fossil-fuel vulnerabilities identified in Chapter 3: gas in buildings and oil in transport.

For heating, policy should support heat pumps where they are technically and economically efficient, particularly in well-insulated buildings and in areas where electric heating can reduce gas demand without creating excessive winter peaks. Building renovation and heat-pump deployment should be planned together. Installing a heat pump in an inefficient building may increase electricity demand unnecessarily, while improving the building envelope first reduces the size and cost of the heating system.

For transport, electric vehicles can reduce oil dependence in road mobility. However, EV policy should not focus only on purchase incentives. Charging infrastructure, distribution-grid capacity, smart charging and tariff design are equally important. If EV charging is unmanaged and concentrated during evening peaks, it can increase system stress. If it is flexible, it can absorb renewable electricity and become a resource for the grid.

Relative prices are crucial. If electricity remains heavily burdened by taxes and network charges while gas remains comparatively cheaper, households may have weak incentives to switch even when electrification is socially desirable. A credible electrification strategy should gradually rebalance tariffs so that electricity prices better reflect decarbonisation and energy-security objectives, while protecting vulnerable consumers from excessive bills.

## 11.4. Treat grids, storage and demand response as strategic infrastructure

The fourth policy implication is that Italy must treat flexibility as strategic infrastructure. Grid reinforcement, storage, interconnections and demand response do not reduce imports as visibly as renewables or heat pumps, but they determine whether those measures can scale.

Transmission investment is essential because renewable potential and electricity demand are not always located in the same places. Distribution networks are equally important because electrification, rooftop solar, heat pumps and EV charging create new local constraints. Policy should therefore accelerate network planning, digitalisation and coordination between transmission and distribution system operators.

Storage should be supported where it creates system value. Batteries are useful for short-duration balancing, solar shifting, congestion management and ancillary services. Pumped hydro and other long-duration options may be more relevant for longer balancing needs. Policy should avoid measuring success only by installed storage capacity; the relevant criterion is whether storage reduces curtailment, congestion, peak demand and fossil backup.

Demand response should become a central part of Italian electricity policy. Dynamic tariffs, smart meters, automation and aggregators can allow consumers and firms to adjust demand according to system conditions.

Industrial loads, EV charging and heat pumps can all provide flexibility if incentives are correctly designed. This would reduce the need for expensive backup capacity and increase the value of domestic renewables.

Market design must evolve accordingly. A high-renewables electricity system requires prices and markets that reflect time, location, scarcity and flexibility. If the market rewards only energy production but not reliability, capacity, flexibility and location, investment will be inefficient. Italy should therefore strengthen markets for balancing, ancillary services, capacity and demand-side participation.

## 11.5. Use biomethane and hydrogen selectively

The fifth policy implication is that biomethane and hydrogen should be used selectively. They should not be treated as general substitutes for fossil gas or as excuses to delay electrification and efficiency.

Biomethane can reduce imported gas in selected uses and has the advantage of being partly compatible with existing gas infrastructure. However, sustainable feedstocks are limited. Policy should therefore prioritise biomethane where it produces the highest system value, such as specific industrial uses, agricultural integration, waste management and cases where electrification is difficult.

Hydrogen should be targeted even more carefully. Green hydrogen has potential for hard-to-electrify industrial processes, chemical feedstocks, refineries, some heavy transport segments and possibly power-system backup. However, it is expensive, infrastructure-intensive and requires large amounts of renewable electricity. Using green hydrogen in sectors that can be directly electrified would waste scarce renewable electricity and increase system costs.

This implies a clear policy hierarchy. Direct electrification should be preferred where it is efficient. Biomethane and hydrogen should be reserved for residual sectors. Public support should be allocated according to additionality, system value and realistic deployment potential, not according to technological enthusiasm.

## 11.6. Protect affordability and industrial competitiveness

A successful energy-security strategy must also be socially and economically sustainable. If the costs of the transition are perceived as unfair, policy support may weaken. Affordability is therefore not a secondary objective; it is part of energy security.

For households, policy should protect vulnerable consumers while preserving incentives to save energy and shift demand. This means avoiding broad untargeted subsidies to energy consumption, which reduce the incentive to invest in efficiency. Support should instead be targeted to low-income households, building renovation, efficient appliances and access to clean technologies.

For firms, especially energy-intensive industries, policy should reduce exposure to fossil-fuel volatility without weakening competitiveness. Efficiency support, renewable power purchase agreements, electrification of suitable processes and targeted low-carbon gases can all help. However, industrial policy should be selective and conditional: public support should be linked to energy savings, emissions reductions and long-term competitiveness, not simply to compensation for high energy prices.

Public finance also matters. Italy should avoid policy instruments that generate large fiscal costs without proportional energy-security benefits. The best policies are those that leverage private investment, reduce risk, target market failures and create durable improvements in the energy system.

## 11.7. Improve governance and coordination

The final policy implication is institutional. Energy dependence is a systemic problem, so fragmented policy is ineffective. Italy needs coordination between energy policy, industrial policy, transport policy, building policy, regional planning and electricity-market regulation.

Permitting, grid planning, renewable auctions, building renovation, EV charging and industrial decarbonisation should not be treated as separate policy areas. They interact. Renewable deployment without grids creates

bottlenecks. Heat pumps without building efficiency increase peak demand. EVs without smart charging increase system costs. Hydrogen without industrial demand creates infrastructure risk.

Governance should therefore be based on clear priorities, stable rules and measurable outcomes. Policy should monitor not only installed capacity or public expenditure, but also actual reductions in imported fossil fuels, final energy demand, emissions, curtailment, peak demand and vulnerability to price shocks.

## 11.8. Overall policy package

The policy package recommended by this paper can be summarised as follows:

| Priority | Policy direction | Expected contribution to energy dependence |
|---|---|---|
| 1 | Targeted energy efficiency in buildings and industry | Reduces gas, electricity and overall final energy demand |
| 2 | Faster renewable deployment with grid-aware planning | Increases domestic low-carbon electricity supply |
| 3 | Electrification of heating and transport | Replaces imported gas and oil in final uses |
| 4 | Grid reinforcement, storage, interconnections and demand response | Enables renewables and electrification to scale reliably |
| 5 | Selective biomethane and hydrogen | Covers residual hard-to-electrify sectors |
| 6 | Tariff reform and affordability protection | Aligns private incentives with social energy-security goals |
| 7 | Better governance and policy coordination | Prevents bottlenecks, duplication and inefficient subsidies |

The main policy conclusion is that Italy should not choose between demand-side and supply-side strategies. It needs both. Demand reduction lowers the size of the dependency problem. Domestic renewable supply reduces exposure to fossil fuel imports. Electrification shifts final uses away from imported gas and oil. Flexibility makes the new electricity system reliable. Renewable gases cover the remaining difficult sectors.

A credible Italian energy-security strategy should therefore be based on sequencing and coordination. In the short term, Italy should prioritise energy efficiency, demand response, permitting reform and renewable deployment. In the medium term, it should accelerate heat pumps, EVs, grid upgrades and storage. In the long term, it should develop targeted biomethane and hydrogen solutions for hard-to-electrify sectors.

The final implication is that energy security and decarbonisation are not separate objectives. If properly designed, the energy transition can reduce both emissions and import dependence. The challenge is not technological availability alone, but policy design: incentives, regulation, infrastructure planning and market rules must work together. This is where Italy’s energy dependence can be reduced most effectively.

# *12. Conclusions*

This paper analysed how Italy can reduce its energy dependence while balancing energy security, environmental sustainability and economic affordability. The starting point was that Italy's vulnerability is not simply the result of importing a high share of energy. It is the outcome of a specific energy-system structure: oil dependence in transport, gas dependence in buildings and industry, gas-fired electricity generation, electricity imports and infrastructure constraints. For this reason, reducing energy dependence requires a systemic strategy rather than a single technological solution.

The research question was:

**How can Italy effectively reduce its energy dependence while balancing energy security, environmental sustainability and economic affordability?**

The answer developed in the paper is that Italy should not pursue full energy self-sufficiency, which would be unrealistic and economically inefficient. Instead, the objective should be to reduce systemic vulnerability. This means lowering exposure to imported fossil fuels, reducing sensitivity to international price shocks, increasing domestic renewable production, improving energy efficiency and making the electricity system more flexible and reliable.

The main finding is that the strongest strategy is a coordinated portfolio based on five elements.

First, **energy efficiency** should be the first priority. It reduces the amount of energy required to provide heating, mobility, industrial output and other energy services. In the Italian case, this is particularly important in buildings and industry, where lower demand directly reduces natural gas consumption and price exposure. Efficiency is therefore not only a climate policy, but also an energy-security and affordability policy.

Second, **renewable electricity** is the main supply-side pillar. Solar and wind increase domestic low-carbon production and reduce the role of imported fossil fuels in electricity generation. Their importance grows further when final uses are electrified. However, renewables are not sufficient on their own. Their value depends on the ability of the system to absorb, transport and balance variable generation.

Third, **electrification** is essential for replacing imported fossil fuels in final uses. Heat pumps can reduce gas demand in buildings, while electric vehicles can reduce oil dependence in transport. However, electrification is conditional. It reduces energy dependence only if the additional electricity demand is increasingly supplied by domestic renewables and if the grid can manage new load profiles.

Fourth, **grid flexibility, storage, interconnections and demand response** are enabling conditions. They do not reduce imports as directly as efficiency or renewables, but they determine whether a more renewable and electrified system can function reliably. Without flexibility, Italy may install renewable capacity but continue to rely on gas-fired generation and imports during critical hours. With flexibility, renewable electricity becomes a real source of energy security.

Fifth, **biomethane and hydrogen** should be used selectively. They can contribute to reducing import dependence in hard-to-electrify sectors, but they should not be treated as universal substitutes for gas or oil. Biomethane is limited by sustainable feedstock availability, while green hydrogen is limited by cost, infrastructure needs and conversion losses. Their highest value is in specific industrial processes, chemical uses, heavy transport niches and backup applications where direct electrification is difficult.

The comparative assessment showed that these options are mostly complements, not substitutes. Efficiency reduces the size of final demand. Electrification changes the energy carriers used in final consumption. Renewables provide domestic low-carbon electricity. Flexibility makes that electricity reliable and valuable. Biomethane and hydrogen address residual sectors. The policy challenge is therefore not to choose one technology, but to coordinate them in the right sequence.

The sequence matters. In the short term, Italy should prioritise energy efficiency, demand response, renewable permitting reform and better use of existing infrastructure. In the medium term, it should accelerate renewable deployment, heat pumps, electric vehicles, storage and grid upgrades. In the long term, it should develop targeted solutions for hard-to-electrify sectors, including biomethane and green hydrogen where they create genuine system value.

The analysis also confirms the relevance of the energy trilemma. A strategy focused only on security may become too costly. A strategy focused only on decarbonisation may create reliability problems if infrastructure is not ready. A strategy focused only on affordability may delay the investments needed to reduce long-term vulnerability. The best policy mix is therefore the one that reduces imports, emissions and price exposure while preserving competitiveness and social acceptance.

The main policy implication is that Italy needs coordinated governance. Energy dependence cannot be reduced through fragmented incentives. Building renovation, heat pumps, electric vehicles, renewable auctions, grid planning, storage, interconnections and industrial decarbonisation are interconnected. If one element moves without the others, the transition becomes slower, more expensive and less effective.

This paper has some limitations. The analysis is qualitative and based on a multi-criteria framework rather than on a full quantitative energy-system model. It does not estimate exact avoided imports, investment costs or emissions reductions for each option. Future research could extend the analysis by modelling alternative scenarios to 2030 and 2050, comparing different electrification pathways, or estimating the fiscal and distributional impacts of specific policy packages.

Despite these limitations, the conclusion is clear. Reducing Italy's energy dependence does not mean isolating the country from European energy markets or replacing every imported unit of energy with domestic production. It means building a less vulnerable energy system. The most credible pathway is to consume less energy, electrify the uses that can be electrified efficiently, produce more domestic renewable electricity, strengthen flexibility and reserve renewable gases for the sectors where they are truly needed.

In this perspective, energy security and decarbonisation are not separate objectives. If policies are well designed, the energy transition can reduce both emissions and import dependence. The key condition is coordination. Italy's energy dependence will not be solved by one technology, one subsidy or one infrastructure project. It will be reduced by a coherent portfolio of demand reduction, domestic renewable supply, electrification, flexibility and targeted low-carbon fuels.

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